

BIOLOGY GRADE 11: ECOLOGY

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.2 (9 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	11
Subject	Biology
Strand	Strand 1.0: Cell Biology and Biodiversity
Sub-Strand	Sub-Strand 1.2: Ecology
Total Duration	9 lessons × 40 minutes = 360 minutes total
Sub-Strand Content	– Ecosystems (forest, grassland, desert and aquatic) and ecological terminologies (habitat, niche, population, community, biome, ecosystem, biomass, carrying capacity) – Adaptations of organisms to different habitats (xerophytes, mesophytes, hydrophytes, halophytes, desert and aquatic animals) – Characteristics of ecosystems and of populations (density, dispersion, growth) – Estimating population size (direct count and sampling methods: line transect, belt transect, quadrat, capture-recapture) – Factors that affect populations in ecosystems (biotic and abiotic factors) – Nutrient cycles and their importance in ecosystems (nitrogen and carbon cycles) – Effects of environmental pollution (including global warming and climate change)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - describe the characteristics of different ecosystems in nature, - explain the adaptations of organisms to different habitats, - describe factors affecting population sizes in ecosystems, - estimate population sizes of organisms in ecosystems, - illustrate the nitrogen and carbon cycles in ecosystems, - describe the effects of pollution in the environment, - appreciate the link between environmental pollution and climate change.
Core Competencies	– Citizenship: The learner solves problems in the local community as they make posters to sensitise the community about the effects of environmental pollution. – Self-efficacy: The learner communicates clearly with others while brainstorming on the causes and impacts of climate change. – Communication and collaboration: Learners work in groups to sample school-compound populations, share data and present findings. – Critical thinking and problem solving: Learners use evidence to explain why fish are declining in the river and propose restoration actions. – Creativity and imagination: Learners design models of nitrogen and carbon cycles and pollution-awareness posters. – Digital literacy: Learners use Khan Academy readings, PhET simulations and internet research on pollution in Nairobi and Athi Rivers.

Core Values	– Integrity: The learner utilises resources prudently while making charts and models to illustrate nitrogen and carbon cycles. – Patriotism: The learner demonstrates love for the country while sensitising the community about the effects of environmental pollution. – Respect: Learners respect living organisms and the habitats they sample, and one another's views in group work. – Responsibility: Learners return sampled areas undisturbed and take responsibility for caring for local water bodies. – Ethics: Learners record data honestly and handle organisms and counters ethically.
Science & Engineering Practices	– Asking questions and defining problems: Learners generate questions about why fish are declining and add them to the Driving Question Board. – Developing and using models: Learners build and revise a cumulative model of the river ecosystem and construct nitrogen and carbon cycle models. – Planning and carrying out investigations: Learners carry out direct counts, quadrat, belt transect and capture-recapture investigations in the school compound. – Analyzing and interpreting data: Learners calculate population density and estimates, and compare sites. – Using mathematics and computational thinking: Learners apply the capture-recapture formula (marked $\times$ second sample $\div$ recaptured) and density calculations. – Obtaining, evaluating and communicating information: Learners use readings, simulations and research to present findings on pollution. – Engaging in argument from evidence: Learners argue the cause of fish decline using evidence, correcting the idea that fish are simply old or lack food.
Pertinent & Contemporary Issues (PCIs)	– Environmental conservation: Learners study how ecosystems function and propose ways to restore the river ecosystem. – Waste Management: The learner appreciates the importance of proper waste management as they sensitise the community about the effects of environmental pollution and their mitigation measures. – Prevention of Global Warming: The learner appreciates the need to reduce greenhouse gas emissions while observing and discussing the effects of environmental pollution. – Climate change: Learners link pollution and carbon cycle disruption to global warming and climate change. – Health: Learners relate polluted water and unpleasant smells to risks to human and animal health.
Career Connections	– Wildlife biologists: Study populations and use sampling methods such as those practised in the lessons. – Ecological consultants: Assess ecosystems and advise on pollution and habitat impacts. – Conservation scientists: Manage resources such as Lake Victoria fisheries and Kenyan forests. – Park rangers: Protect ecosystems and wildlife in places such as Maasai Mara and Nairobi National Park. – Restoration ecologists: Rehabilitate degraded rivers, wetlands and forests.
Focus for Lessons	This unit is anchored in a local, Kenyan river that is losing its fish while algae, aquatic weeds and bad smells increase. Learners progress from ecological terms and habitat adaptations, through hands-on population estimation in the school compound, to factors affecting populations, nutrient cycles and pollution, steadily building an explanation and community action plan for the river. Each lesson follows Predict, Observe, Explain, Driving Question Board and Model Building phases.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why are fish disappearing from a water body that once supported many fish, and what can the community do to restore the ecosystem? KICD KEY INQUIRY QUESTIONS: 1. How do organisms interact with their environment?

Anchoring Phenomenon	Learners watch a Kenyan video (from 2:20 to 4:50) made by local people showing a nearby river that once had many fish and now yields far fewer. At the same time, people notice increased growth of algae and aquatic weeds, and the water sometimes has an unpleasant smell. It is puzzling because the river is still full of life, yet the fish are disappearing. Learners' initial ideas (the fish are old, or they have no food) do not explain why algae and weeds are booming or why the smell appears, so they must investigate how organisms interact with their environment and with human activity.
Storyline Thread	Lesson 1: Define ecosystems and ecological terms through a Khan Academy reading; ends with the question: How are organisms adapted to live in different habitats? Lesson 2: Classify organisms by habitat (xerophytes, mesophytes, halophytes, hydrophytes, desert and aquatic animals) and study their adaptations through a reading; ends with: What are the characteristics of populations? Lesson 3: Characteristics of populations and direct counting in the school compound (e.g. 10 m × 10 m areas), comparing population density between sites; ends with: Is the distribution of grass uniform throughout the school field? Lesson 4: Quadrat sampling of grass/weed populations using 1 m × 1 m quadrats placed randomly to estimate population size and density; ends with: How does the population of plants change as environmental conditions change across the school compound? Lesson 5: Belt transect across a changing habitat (open field, shaded area, pathway, garden) recording plant abundance at regular intervals; ends with: What factors affect population size in an ecosystem? Lesson 6: Capture-recapture simulation with 100 counters, using estimated population = (marked × second sample) ÷ recaptured; links to estimating animal populations such as fish in the river. Lesson 7: Factors affecting population size (biotic and abiotic) using a Khan Academy reading and a PhET predator and food scarcity simulation; ends with: How do nitrogen and carbon affect living organisms? Lesson 8: Discuss and build models of the nitrogen and carbon cycles in ecosystems, linking nutrient enrichment to algae growth; ends with: Does pollution affect organisms? Lesson 9: Walk to the neighbouring shopping centre to note types of pollution, then research and present on effects of pollution, including garbage rivers and effluent in the Nairobi and Athi Rivers, global warming and climate change, and community solutions to restore the river.
Supporting Phenomena	– Which area of the school compound has the largest population of a particular organism (e.g. grass, weeds, fallen leaves)? – Grass and weeds do not seem to be spread uniformly across the school field, and plants change from open field to shaded area, pathway and garden. – Predator and prey numbers rise and fall in a PhET simulation when food becomes scarce. – Garbage-choked rivers and effluent in the Nairobi and Athi Rivers, and pollution seen at the neighbouring shopping centre.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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LESSON 1 (80 minutes (2 x 40-minute periods)): Where Have the Fish Gone? Launching Our Investigation into Ecosystems

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon of disappearing fish in a Kenyan river, elicit learners' initial ideas, and build the shared ecological vocabulary (ecosystem, habitat, niche, population, community, biome, biomass, carrying capacity, ecological succession) needed to explain it.
Knowledge	By the end of the lesson, the learner should be able to define an ecosystem and the terms used in ecology (habitat, niche, population, community, biome, ecosystem, biomass, carrying capacity, ecological succession) and identify examples of each in a Kenyan river ecosystem.
Skills	Asking questions about a phenomenon; developing initial models (SEP); obtaining, evaluating and communicating information from a reading (SEP); using evidence and observation to record patterns; collaborating in groups.
Attitudes	Curiosity about local environments; integrity in using resources prudently; patriotism and concern for Kenya's rivers and lakes; respect for peers' ideas; appreciation of the link between human activity and environmental pollution.
Key Inquiry Question	How do organisms interact with their environment?
Purpose in Storyline	Lesson 1 opens the unit. Learners watch the Kenyan video of a river where fish are declining while algae, aquatic weeds and bad smell increase. Their initial ideas (fish are old, or have no food) cannot explain these observations, which creates the need to know. A Khan Academy reading gives the vocabulary of ecology. The lesson ends with the question: How are organisms adapted to live in different habitats?
Safety Notes	No hands-on practical in this lesson. Remind learners that when later visiting rivers, dams or lakes (e.g. Nairobi River, Lake Victoria shores) they must go with a teacher, avoid contact with polluted water, and never taste or drink water from streams. Ensure seating allows safe movement during the gallery walk and DQB posting.

B. LESSON OVERVIEW

Learners watch a 2:20-4:50 Kenyan community-made video of a river that once supported many fish but now yields far fewer, while algae and aquatic weeds grow thickly and the water sometimes smells unpleasant. Learners first record individual predictions in their books, then observe the video and a Khan Academy reading on ecosystems and ecological terms. They use the terms to describe the river (habitat, population, community, niche, biotic and abiotic factors). Their initial ideas (fish are old, or lack food) are tested against the evidence and found incomplete. Learners then write questions on sticky notes to open the Driving Question Board, sketch an initial model of the river, and complete an exit ticket. The lesson ends with the question: How are organisms adapted to live in different habitats?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Ecology	Learners sit in groups of 4. Individually, in their exercise books, they answer: 'A river near a	Welcome learners: 'Today we begin an investigation about a real problem in our own country.' Pose	Eliciting prior knowledge through individual prediction and Think-Pair-Share. Writing before talking	Observe whether learners (1) write a prediction with a reason, (2) include both living and non-living

introduction Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Apply: populations, communities, and ecosystems Source: Khan Academy (English - US curriculum) Search ARES for similar readings	village in Kenya (such as the Nyando near Kisumu or a stream feeding Lake Victoria) once gave fishermen many tilapia and now gives very few. What do you think happened to the fish? What might be happening in the river?' They write a prediction and one reason, then draw a quick sketch showing living things and non-living things they think are in the river. They share with a partner and then record their group's top idea on a sticky note for the DQB.	the question on the board and read aloud. Give WAIT TIME of 10-15 seconds of silent thinking before any writing starts. Say: 'Write what you think, even if you are not sure. There are no wrong ideas today. We will come back to these.' Circulate and note 3 common ideas. Cold-call 4 learners (not volunteers) to share their partner's idea, not their own: 'Wanjiru, what did your partner say?' Record ideas on the left side of the board under 'Our first ideas'. Do not correct yet. Ask: 'What would we need to see to know whether this idea is true?'	lets every learner commit to an idea, so later evidence can be compared with first thinking. Sketch modelling (SEP: Developing and Using Models) shows how learners currently picture the river system.	components in the sketch, (3) use everyday ideas such as 'old fish', 'no food', 'overfishing' or 'dirty water'. Look-fors: at least 80% of learners have a written prediction in books. Tally common ideas on the board to guide later teaching.
Observe Phase VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Habitat and Niche (Flexbook) Source: CK-12 Search ARES for similar readings	Part A (Video): Learners watch the Kenyan community video from 2:20 to 4:50 showing a river that once had many fish and now yields far fewer, with thick algae, aquatic weeds and a sometimes unpleasant smell. In a 3-column table (What I see What I hear What I wonder) they record observations. Part B (Reading): Learners read the Khan Academy reading on ecosystems and ecological terms, individually then in groups. They highlight and list definitions of ecosystem, habitat, niche, population, community, biome, biomass, carrying capacity and ecological succession. In groups, they complete a table matching each term to an example from the river in the video (e.g. population: all tilapia in the river; community: fish, algae, weeds, insects, frogs, birds; habitat: the river water and banks; abiotic factors: water temperature, oxygen, light, smell/pollutants).	Before playing: 'Watch carefully. Do not talk. Write only what you see and hear.' Play 2:20-4:50 once without pause, then replay and pause at key moments (fish catch, algae, weeds, smell). Ask: 'What is living in this river? What is not living?' WAIT TIME of 10-15 seconds after each question. Cold-call 3 learners to share one observation each. Press for evidence: 'What did you see that makes you say that?' Introduce the reading: 'The scientists' words in this reading will help us describe the river exactly.' Distribute or direct learners to the Khan Academy reading on ecosystems (or printed copies for the offline platform). Give 10 minutes for silent reading and highlighting. Circulate; ask 2 groups: 'Which term describes all the tilapia in this river together? Which describes all the living things together?' Clarify misconceptions, especially habitat vs niche and population vs community. Use Kenyan examples: 'The Maasai Mara wildebeest are a population; the	Structured observation with an 'I see / I hear / I wonder' notice-and-wonder routine, followed by an annotated reading (SEP: Obtaining, Evaluating and Communicating Information). Matching vocabulary to the real river anchors abstract terms in the phenomenon and builds a shared language for later lessons.	Check that observation tables contain specific details (algae, weeds, smell, fewer fish in nets) rather than general statements. Check term-example tables: learners correctly distinguish population (one species) from community (many species) and habitat (where) from niche (role). Look-fors: at least 7 of 9 terms correctly matched to river examples in each group; highlighted definitions in own words.

		Mara ecosystem is the whole system with the grass, lions and rainfall.'		
Explain Phase  VIDEO: Ecology introduction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Habitat and Niche Practice Source: CK-12  Search ARES for similar readings	Using the vocabulary, learners return to their predictions. In groups, they write a claim-evidence-reasoning (CER) paragraph: 'Which of our first ideas (fish are old; fish have no food) can the video and the reading explain, and which can they not?' They use terms such as ecosystem, community, biotic and abiotic factors, carrying capacity and biomass. They explain why old age or lack of food alone cannot account for the booming algae and weeds and the smell. They then each write a one-sentence tentative explanation of how living things and the environment, including people's activities (e.g. farm run-off, sewage, washing, waste), might be interacting in the river. Groups share in a short whole-class discussion.	Ask: 'Look at our first ideas on the board. If the fish are simply old, what would we expect to see in the algae and the smell? Does it fit?' WAIT TIME of 10-15 seconds. Cold-call 4 groups, one learner each, to share a claim with evidence. Prompt: 'Use the word ecosystem or community in your explanation.' Probe: 'Who can add to or challenge that idea?' Highlight that the ideas explain part of the evidence only. Say: 'Some of our ideas are good starting points but they leave questions. Which observation is still unexplained?' Record unexplained observations on the board (algae and weeds booming, smell). Do not give the full answer on eutrophication yet; say: 'We will find out in coming lessons.'	Claim-Evidence-Reasoning (SEP: Constructing Explanations) and consensus discussion. Comparing first ideas with evidence helps learners see gaps in their thinking, which motivates the investigation of how organisms interact with their environment and human activity.	Listen for learners who (1) use at least 2 ecological terms correctly, (2) cite specific evidence from the video, (3) admit that some observations remain unexplained. Look-fors: each group's CER paragraph contains a claim, at least one piece of evidence and a link between them. Note learners who still think the fish problem is only about fish and not the whole ecosystem.
Driving Question Board (DQB) Creation  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Apply: populations, communities, and ecosystems Source: Khan	Learners read the Driving Question: 'Why are fish disappearing from a water body that once supported many fish, and what can the community do to restore the ecosystem?' Individually, they write 2 questions on sticky notes (e.g. 'Why does algae grow so much?', 'Where does the smell come from?', 'Do the weeds harm the fish?', 'How do fish live in water?'). In groups they sort questions into categories (causes, organisms, human activities, solutions). Each group posts its best 3 questions on the Driving Question Board and one group member explains them. Learners also copy their initial	Display the DQB chart paper with the driving question at the top. Say: 'A good question helps us investigate. Write questions you really want answered.' Model one example: 'How do algae and weeds affect fish in the river?' Give 3 minutes of silent writing (WAIT TIME 10-15 seconds before sharing). Ask groups to cluster questions. Cold-call 5 learners (one per group) to place and explain a question: 'Why did your group place this here?' Identify and group questions that connect to the next lesson: 'Several of you asked how fish live in water and how plants survive in a river. That leads us to our next question.'	Driving Question Board: learners' own questions become the path of the unit and show them that their thinking drives learning (SEP: Asking Questions and Defining Problems). Grouping questions reveals patterns and builds ownership of the investigation.	Check that every learner contributed at least 2 questions and that questions are investigable rather than yes/no. Look-fors: at least 3 categories emerge on the DQB; some questions link to adaptations, food, pollution or the community's role. Use the sorted questions to plan upcoming lessons.

Academy (English - US curriculum) Search ARES for similar readings	ideas into their books for later comparison.	Photograph or label the DQB and keep it visible in the classroom (or a notebook copy for the offline platform).		
Model Building Phase VIDEO: Worked example: Logistic model word problem <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Habitat and Niche (Flexbook) Source: CK-12 Search ARES for similar readings	Individually, learners draw an initial model of the river ecosystem on a page in their books. It must show: biotic components (fish such as tilapia and Nile perch, algae, aquatic weeds, insects, birds), abiotic components (water, sunlight, oxygen, temperature), and arrows showing how they interact, with a label for possible human activities (farming, washing, sewage, waste dumping, fishing). They label at least 4 ecological terms on the model (e.g. population, community, habitat, ecosystem) and add a box titled 'What I still wonder'. Then they complete the Exit Ticket: (1) Define ecosystem in your own words. (2) Give one example each of habitat and population from the river. (3) Write one question you still have. Finally, the teacher poses the closing question: 'How are organisms adapted to live in different habitats?'	Say: 'Draw your best current picture of the river. It will change as we learn.' Provide a quick example label (e.g. 'algae = producers') but not a full model. WAIT TIME of 10-15 seconds before drawing starts. Circulate and ask 3 learners: 'What does this arrow show?' Distribute exit tickets (or ask learners to write on a half sheet). Collect them at the door. Close: 'Fish live in water, but weeds, cacti and camels live elsewhere. How are organisms adapted to live in different habitats? That is our question for the next lesson.' Remind them to keep their models and predictions for later comparison.	Exit ticket and initial model. Sketch modelling (SEP: Developing and Using Models) externalises current thinking so it can be revised after each lesson. The exit ticket gives each learner a private way to show understanding and raise remaining questions.	Exit ticket. Look-fors: (1) definition mentions living and non-living components interacting in an area; (2) habitat given as a place (e.g. the river water) and population as one species in an area (e.g. all tilapia); (3) one question linked to the phenomenon. Sort tickets into 'got it', 'partly', 'not yet' piles and begin the next lesson with a quick re-teach for the 'not yet' group. Models should show at least 3 biotic and 2 abiotic components and arrows.

D. TEACHER REFLECTION

1. Which first ideas did learners hold most strongly (e.g. 'fish are old', 'no food', 'overfishing'), and how did the video and reading challenge them? 2. Did learners confuse habitat with niche or population with community? Which examples worked best? 3. Did every learner contribute questions to the DQB? Which learners need more support to speak or write? 4. How well did the Kenyan context (rivers near Kisumu, Lake Victoria, Nairobi River) connect to learners' lives? 5. What should be adjusted before Lesson 2 on adaptations, based on the exit ticket results?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	A Kenyan river that once had many fish now yields few, while algae and aquatic weeds grow thickly and the water sometimes smells unpleasant. The river is still full of life.
What did I learn?	An ecosystem is a community of living organisms interacting with each other and with non-living factors in an area. Ecological

	terms such as habitat, niche, population, community, biome, biomass, carrying capacity and ecological succession help describe how the river works.
How does this explain the phenomenon?	Fish being old or lacking food cannot alone explain the booming algae, weeds and smell. The river is an ecosystem where organisms and the environment, including human activity, interact, so changes in one part can affect others. We still need to learn how organisms are adapted to their habitats and what causes these changes.

LESSON 2 (40 minutes (single lesson; extend to 80 minutes if a double period is available)): Who Lives Where? Habitat Classification and Adaptations in a Kenyan River and Beyond

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners classify organisms by the habitat in which they live (xerophytes, mesophytes, hydrophytes, halophytes, desert animals and aquatic animals) and explain how structural and physiological adaptations help them survive. They then use this to explain why algae and aquatic weeds thrive while fish decline in the river.
Knowledge	By the end of the lesson, the learner should be able to: (b) explain the adaptations of organisms to different habitats (xerophytes, mesophytes, hydrophytes, halophytes, desert and aquatic animals). Learners know that organisms are grouped by habitat and that each group has adaptations suited to water availability, salinity, temperature and oxygen.
Skills	Developing and using models (labelled sketches of adaptations); obtaining, evaluating and communicating information (jigsaw reading and teaching peers); constructing explanations linking adaptations to the river phenomenon; communicating clearly.
Attitudes	Core competencies: Self-efficacy (the learner communicates clearly with others while explaining and teaching adaptations). Citizenship (the learner solves problems in the local community by linking habitat knowledge to the river problem). Values: Integrity (utilises resources prudently, sharing one reading per group). Patriotism (shows love for the country by appreciating Kenya's diverse habitats from Lake Victoria to Chalbi Desert). PCI: Waste Management and Prevention of Global Warming (appreciating why healthy habitats need protection).
Key Inquiry Question	How do organisms interact with their environment?
Purpose in Storyline	Lesson 1 raised the puzzle of the river that once had many fish but now has algae, weeds and a bad smell. Lesson 2 gives learners the vocabulary and evidence to see that organisms are matched to habitats through adaptations: aquatic plants such as water hyacinth thrive in nutrient-rich water, while fish with gills depend on dissolved oxygen. This leads to the next question: What are the characteristics of populations?
Safety Notes	This is a reading-based lesson with no hazardous materials. If learners sketch or handle plant specimens (e.g. cactus or Euphorbia/candelabra spines, sisal), warn about sharp spines and thorns; handle with paper or gloves and wash hands. If a field walk to a nearby stream or pond is added, learners must stay with the teacher, keep away from the water's edge, and not touch algae, weeds or smelly water with bare hands.

B. LESSON OVERVIEW

Learners start by predicting which Kenyan organisms live where and what features help them. They then do a jigsaw reading in expert groups on six habitat groups: xerophytes, mesophytes, hydrophytes, halophytes, desert animals and aquatic animals. They return to home groups to teach peers and complete a summary table. They use the evidence to explain why the river favours algae and aquatic weeds over fish (the weeds are hydrophytes; algae are protists, not hydrophyte plants), add new questions to the Driving Question Board, and revise their emerging model of the river ecosystem. The lesson ends with oral questions and the question: What are the characteristics of populations?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Matter and energy in food webs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Habitat Destruction (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners recall the river video (2:20 to 4:50) and look at six quick prompts on the board: a cactus-like Euphorbia in Turkana, a maize plant in Kakamega, water hyacinth on Lake Victoria, mangrove at Mombasa (Mida Creek), a camel in Marsabit/Chalbi, and tilapia in the river. Individually they write for 2 minutes: 'Which habitat does each live in, and what feature helps it survive there?' Then they pair up and share. They then predict: 'Why might algae and weeds do well in the river while fish do not?'	Display the six prompts and say: 'Remember our river where fish are disappearing. Before we read, I want your best thinking. There are no wrong predictions today.' Give 2 minutes of silent writing. Say: 'Turn to your partner and share.' Cold-call 4 learners, with WAIT TIME of 10-15 seconds after asking: 'What feature of the camel or the mangrove helps it live where it does?' Record predictions on the board under 'Our first ideas' without correcting. Say: 'Hold on to these. Our reading will test them.'	Prediction with Think-Pair-Share: writing before talking makes prior ideas visible and commits learners to a claim, so later evidence can confirm or revise it. Connecting adaptations to the river phenomenon primes learners to see that organisms are matched to their environments.	Observe whether learners (a) name a habitat for at least 4 of 6 organisms, (b) mention any structural feature (spines, thick stems, gills, broad leaves, salt glands, humps). Tally on a class list how many learners link the camel/cactus to water conservation. Look for misconceptions such as 'plants in deserts do not need water' or 'algae grow because fish are old'. Note these for the Explain Phase.
Observe Phase  VIDEO: Matter and energy in food webs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Habitat and Niche (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	JIGSAW, Round 1 (Expert groups, 12 minutes): Learners are in 6 expert groups, each reading the assigned section of 'read here' on one habitat group: (1) xerophytes (e.g. Euphorbia, sisal, Opuntia, acacia in Turkana/Kajiado), (2) mesophytes (e.g. maize, beans, sukuma wiki in Kakamega/Nakuru), (3) hydrophytes (e.g. water hyacinth, water lily, papyrus in Lake Victoria/Naivasha), (4) halophytes (e.g. mangroves at Mombasa/Lamu, salt-tolerant plants at Lake Magadi), (5) desert animals (e.g. camel, gerbil, fennec fox, kangaroo rat-like rodents), (6) aquatic animals (e.g. tilapia, Nile perch, omena/dagaa in Lake Victoria). In each group, they identify the habitat conditions, list	Distribute the reading (one copy per expert group; learners share to use resources prudently). Say: 'You are now the class expert on your habitat. Every person must be ready to teach it.' Give task cards: 'List 3 conditions of the habitat, 3 adaptations, and 1 link to our river.' Circulate and ask: 'Which part of the organism helps it get or keep water/oxygen?' and 'What would happen if this organism was moved to the river?' Wait 10-15 seconds after each question before prompting. Check that each member has notes. Say: 'You have 2 minutes left; put the key points in your own words.' Cold-call 2 members per group to confirm readiness before moving to Round 2.	Jigsaw (Round 1: Expert groups): Dividing a dense reading into parts lets each learner become responsible for deep understanding of one habitat group. Individual accountability and peer support build comprehension and ensure every learner contributes evidence for the whole class.	Look for: each expert group produces a labelled sketch, 3 correct adaptations and a river link; every member can explain the notes without reading verbatim. Indicators of success: learners link features to function (e.g. 'thick waxy cuticle reduces water loss', 'air spaces in water hyacinth leaves help it float', 'gills extract dissolved oxygen'). Note groups that list features without function and prompt: 'How does this help it survive?'

	structural and physiological adaptations, draw one labelled sketch, and prepare a 2-minute teaching summary. They also note one link to the river phenomenon, e.g. hydrophytes and nutrients, or fish and oxygen.			
Explain Phase  VIDEO: Matter and energy in food webs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Habitat and Niche Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	JIGSAW, Round 2 (Home groups, 12 minutes): Learners regroup into mixed home groups of 6 (one expert from each habitat). Each expert teaches for 2 minutes while others complete a summary table: Habitat group Conditions Adaptations Kenyan example. As a group, they then answer: 'Why do algae and aquatic weeds (hydrophytes) grow so well in the river while fish decline? What conditions in the river are helping one group and harming another?' They write a 3-sentence claim-evidence-reasoning explanation, e.g. 'Hydrophytes and algae thrive when water is rich in nutrients and warm; excess growth uses up dissolved oxygen when it decays, which causes smell and harms fish that rely on gills.'	Say: 'Now you teach each other. Listen carefully; you will be tested on all six.' Time each expert at 2 minutes (call 'Switch'). Circulate and listen for accuracy. Pose the whole-group question: 'Which organisms in our river benefit from the changes, and which do not?' WAIT TIME of 10-15 seconds. Cold-call 5 learners from different groups to share claims and ask 'What evidence from the reading supports that?' Address misconceptions: 'Adaptations are not chosen by the organism; they arise over generations through natural selection.' Clarify that the unpleasant smell and the weeds are signs of an environment changing, and that we still need to find out why (next lessons).	Jigsaw (Round 2: Teaching and synthesis) combined with Claim-Evidence-Reasoning: Teaching peers consolidates understanding, and applying habitat/adaptation ideas to the river phenomenon moves learners from description to explanation of interactions between organisms and environment.	ORAL QUESTIONS (observable): 'What is the difference between a xerophyte and a hydrophyte?' 'Give one adaptation of a halophyte and say how it helps.' 'Why can a tilapia not live in a desert?' 'How could excess algae affect fish?' Success indicators: at least 80% of cold-called learners give an accurate adaptation with its function and a Kenyan example; groups correctly complete all six rows of the table. Record names of learners needing support and revisit during DQB time.
Driving Question Board (DQB) Creation  VIDEO: Matter and energy in food webs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML:	Learners add sticky notes to the class Driving Question Board under the central question 'Why are fish disappearing from a water body that once supported many fish, and what can the community do to restore the ecosystem?' They write questions and ideas that arose, such as 'Why do water hyacinth and algae grow so fast in Lake Victoria?', 'What do fish need to survive in the water?', 'How many fish are in the river?', 'What causes the smell?'. They sort questions into clusters (adaptations, populations, nutrients/pollution, community	Say: 'Take 2 minutes to write at least one new question from today's reading.' Move learners to the board in small groups. Ask: 'Which questions can we now partly answer, and which do we still need evidence for?' WAIT TIME of 10-15 seconds. Cold-call 3 learners to read and place their questions. Highlight the cluster 'How many fish are left?' and say: 'To know if fish are really disappearing, we need to understand populations. That is our next lesson.' Record any new vocabulary (habitat, adaptation, hydrophyte) on the word wall.	Driving Question Board: Visually tracking questions makes learning purposeful, shows how lesson evidence connects to the phenomenon, and surfaces gaps that motivate the next investigation.	Check that questions are investigable and linked to the phenomenon (e.g. 'How does algae affect oxygen for fish?' rather than 'What is a fish?'). Observe whether learners can state which question their reading helped answer. Note at least 5 new questions about populations, nutrients or pollution for planning Lessons 3-9.

Habitat Destruction (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	action) and place a green tick on questions partly answered by today's reading.			
Model Building Phase  VIDEO: Matter and energy in food webs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Habitat and Niche (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners start a simple model of the river ecosystem in their notebooks (a labelled sketch of the river, fish, algae, weeds, riverbank plants, people and arrows). They add two new elements from today: (1) labels showing which organisms are hydrophytes/aquatic animals and an adaptation of each, and (2) a short note 'Organisms are adapted to specific habitat conditions; changes in conditions (e.g. nutrients, oxygen) favour some organisms and harm others.' They add a question mark to show what is still unknown (how many organisms there are, why conditions changed). They compare with a partner and add one improvement.	Say: 'Draw our river as a system. Show which organisms live there and which conditions they need.' Display an example arrow: 'Nutrients → algae → less oxygen → fish?' and say: 'This is only a possible link; mark it with a question mark until we have evidence.' Circulate and ask: 'Which adaptation explains why this organism is in this part of the river?' WAIT TIME of 10-15 seconds. Cold-call 3 learners to share one model addition. End with oral questions: 'Name one adaptation of an aquatic animal and one of a hydrophyte.' Then say: 'Today we learned how organisms are matched to habitats. But how many organisms live in the river, and how do we describe a group of the same species? What are the characteristics of populations?'	Cumulative Model Building: Revising a visible model after each lesson helps learners integrate new evidence into a coherent explanation and see where gaps remain. Connecting adaptations to ecosystem conditions builds the idea of interactions between organisms and environment.	Look for models that (a) include at least 3 organisms with habitat labels, (b) show at least 2 adaptations with their functions, (c) include a question mark or unresolved question. Indicator of understanding: learners can say, 'Hydrophytes and algae thrive where nutrients are abundant, while fish need oxygen.' ORAL QUESTIONS to the class as a quick check: 'What is an adaptation?' 'Name a Kenyan halophyte.' Collect 5 models for review.

D. TEACHER REFLECTION

After the lesson, reflect: (1) Did every expert group correctly interpret their reading, and which habitat group (e.g. halophytes) caused the most difficulty? (2) Which learners confused structural and physiological adaptations, or used phrases such as 'the plant wanted to'? How will you address this in Lesson 3? (3) Were learners able to connect hydrophytes/algae growth to the river phenomenon, or did they treat adaptations as unrelated facts? (4) Did the jigsaw timing work, and were all learners involved in teaching? (5) Which new DQB questions about populations can drive Lesson 3? Plan to revisit misconceptions with Kenyan examples (e.g. camel humps store fat, not water).

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We read about xerophytes, mesophytes, hydrophytes, halophytes, desert animals and aquatic animals and noted the conditions of each habitat and the adaptations of Kenyan organisms such as Euphorbia, maize, water hyacinth, mangroves, camels and tilapia.
What did I learn?	Organisms are classified according to the habitat in which they live, and each group has structural and physiological adaptations (e.g. waxy cuticle and spines, air spaces, salt glands, gills, water-conserving kidneys) that help it survive water, salt, oxygen and temperature conditions.
How does this explain the phenomenon?	In the river, conditions such as extra nutrients and warm water favour algae and aquatic weeds (hydrophytes), while fish with gills need enough dissolved oxygen, so changes in the environment can help some organisms and harm others. We still need to find out how many organisms live there and what the characteristics of populations are.

LESSON 3 (40 minutes): Counting Life: Characteristics of Populations and Direct Counting in the School Compound

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate the characteristics of a population and use direct counting in two 10 m x 10 m areas of the school compound to calculate and compare population density, so they can later judge whether the fish population in the Kenyan river has really declined.
Knowledge	By the end of the lesson, the learner should be able to: (a) define population and population density; (b) describe direct counting as a method of estimating population size and state when it is suitable (small populations whose individuals are easily identified and counted); (c) calculate population density as number of individuals divided by area sampled; (d) explain why populations differ between sites.
Skills	Planning and carrying out investigations (NGSS SEP 3): marking a defined area and counting every individual. Analyzing and interpreting data (SEP 4): recording number, area and density, and comparing sites. Using mathematics and computational thinking (SEP 5): calculating density. Constructing explanations (SEP 6): using counts to explain differences between sites.
Attitudes	Citizenship: solves problems in the local community by applying population counts to the river problem. Integrity: uses resources prudently and records honest counts. Patriotism: shows care for the Kenyan environment. Self-efficacy: communicates clearly within the jigsaw groups.
Key Inquiry Question	How do organisms interact with their environment?
Purpose in Storyline	Lessons 1 and 2 showed that organisms are adapted to their habitats and that ecological terms describe how living things relate to their surroundings. Learners now ask how many individuals live in a place. This lesson gives them the first tool for measuring a population, which they need to test the claim that fish are fewer. It ends with the new question: Is the distribution of grass uniform throughout the school field? This prepares learners for sampling methods (quadrat, transect) in the next lessons.
Safety Notes	Learners stay within the marked school areas and wear closed shoes. Check for snakes, thorny plants or broken glass before counting. Do not uproot or trample plants unnecessarily. Avoid working in the midday sun, so carry water and wear hats. Wash hands afterwards. Each group should have a supervised area in view of the teacher.

B. LESSON OVERVIEW

Learners recall the video of the Kenyan river where fish are fewer. They predict how scientists could know whether fish numbers have truly fallen, and which part of the school compound has the most of one organism (such as grass tufts, weeds or fallen acacia/jacaranda leaves). In jigsaw groups, each learner becomes an expert on one part of the task: marking the area, counting, calculating density, or comparing sites. Experts count organisms in two 10 m x 10 m areas, calculate density, and teach their home group. The class explains when direct counting is suitable and why it would be hard in a river. A Driving Question Board update and a model revision follow, and an exit ticket checks understanding. The lesson closes with the question: Is the distribution of grass uniform throughout the school field?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	(5 min) Learners recall the river video (2:20 to 4:50) in which	Say: 'Remember the river where people said, There used to be	Prediction with partner talk (Think-Write-Pair-Share): committing to a	Observe whether learners (1) write a prediction with a reason, (2) use

 VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Population Size, Density, and Distribution (Flexbook) Source: CK-12  Search ARES for similar readings	people said the fish were fewer. Individually they write a prediction in their notebooks: 'Which area of our school compound has the largest population of grass plants (or weeds), and how could we know for sure instead of just guessing?' They sketch the school compound and mark where they think the organism is most abundant (for example, near the classrooms, under the trees, or by the football field). They then share with a partner and link it to the river: 'How would villagers around the river know that fish have really decreased?'	many fish. Is that a fact or an opinion? How could we turn it into evidence?' Give WAIT TIME of 10-15 seconds before taking any answers. Cold-call 3 learners to share predictions and ask each: 'What makes you think so?' Record predictions on the board under 'Our Predictions'. Introduce the word 'population' by asking: 'What do we call all the fish of one species in the river?' Wait 10 seconds and cold-call 2 learners. Do not correct predictions yet and say: 'Let us test them with real counts.'	written prediction exposes prior ideas, such as guessing or eyeballing, and creates a need for a reliable counting method. Linking to the river phenomenon shows why measurement matters in real community problems.	the word 'population' correctly, and (3) suggest counting or measuring instead of guessing. Note on a checklist which learners say 'look and see' only, and which suggest counting. Aim for at least 20 of 30 learners to suggest counting.
Observe Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Population Size, Density, and Distribution Practice Source: CK-12  Search ARES for similar readings	(15 min) Jigsaw, stage 1 and 2. Class forms home groups of 4, and each member takes a role to become an expert: (1) Area Marker (marks a 10 m x 10 m area using pegs/sticks and a string or measuring tape); (2) Counter (counts every individual of the chosen organism, e.g. grass tufts, dandelions, a weed such as Bidens pilosa (black jack), or fallen leaves of a tree, using a tally); (3) Recorder (writes number of organisms, area sampled and site); (4) Calculator (finds density = number of organisms / area in m²). Expert groups first meet briefly to agree on how to do their job (for example, define what counts as one individual). Then home groups go to Site A (for example, near the classroom block) and mark a 10 m x 10 m area, count all individuals, and record. They repeat at Site B (for example, under a tree or near the football field). Data table: Site Area sampled (m²) Number of organisms Population density (per m²) Notes (shade, soil,	Before going out, demonstrate marking a 10 m x 10 m area using four pegs and say: 'Everyone must agree on what counts as ONE plant before counting, so that we all count the same way.' Circulate to each group and ask: 'How are you making sure you do not count the same plant twice?' and 'What is your area in square metres?' Give WAIT TIME of 10-15 seconds after each question. Cold-call 2 groups to explain their counting rule. Prompt struggling groups: 'What can you do to be sure nothing is missed?' (for example, count in strips). Remind learners of safety. Call the groups back after 12 minutes and ask each Calculator to put the density on the board.	Jigsaw (as requested for Lessons 3 to 6): each learner holds one expert piece of the investigation, and the group must combine them to get complete data. This builds interdependence and shared accountability, and helps learners see that counting, area, and density work together as one idea (population density).	Observable indicators: (1) the area is correctly marked at 10 m x 10 m; (2) the group has a clear counting rule and uses a tally; (3) the table is complete with number, area, and density for two sites; (4) density is calculated correctly (for example, 250 plants / 100 m² = 2.5 per m²). Use a checklist as you circulate and note groups needing support with units or division.

	trampling).			
Explain Phase  VIDEO: 6a-c. Population ecology <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Population Size, Density, and Distribution (Flexbook) Source: CK-12  Search ARES for similar readings	(10 min) Jigsaw, stage 3. In their home groups, each expert teaches the others what they learned: how the area was marked, how counting was done, how density was calculated, and how the sites differ. Groups then answer in writing: (1) Which site had the largest population? Which had the highest density? (2) Suggest reasons for the difference (sunlight, soil, water, trampling by learners, shade from a tree). (3) Why was direct counting possible here? (4) Would direct counting work for the fish in a river or for the algae? Why or why not? Learners write a claim-evidence-reasoning statement: 'Direct counting is suitable when the population is relatively small and organisms can be easily identified and counted.'	Ask: 'What is the difference between population size and population density?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners. Press for evidence: 'Which numbers in your table support your claim?' Probe the river link: 'If fish swim, hide in deep water and cannot be seen, can we count every one? What could we do instead?' (Preview that sampling methods will be used next.) Cold-call 2 groups to share their claim-evidence-reasoning. Correct the misconception that more individuals always means higher density by showing two sites with different areas. Write on the board: 'Population density = number of individuals / area'.	Jigsaw teaching-back with Claim-Evidence-Reasoning: peer teaching consolidates the method and vocabulary, while CER makes learners justify conclusions using their own data. Explaining the limits of direct counting prepares them to see why other methods are needed for mobile or numerous organisms such as river fish.	Look for: (1) correct use of the terms population and population density; (2) a claim supported by actual numbers from their table; (3) at least one sound reason for the site differences; (4) recognition that direct counting is unsuitable for large, hidden or moving populations. Note learners who confuse number with density for follow-up.
Driving Question Board (DQB) Creation  VIDEO: Population regulation <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Population Size, Density, and Distribution Practice Source: CK-12  Search ARES for similar readings	(5 min) Learners return to the Driving Question Board with the question: 'Why are fish disappearing from a water body that once supported many fish, and what can the community do to restore the ecosystem?' Each group writes one sticky note stating what they now know (for example, 'We can count a population and calculate density to compare places') and one new question (for example, 'How can we count fish we cannot see?' or 'Is the grass spread evenly across the whole field?'). They place notes on the DQB under 'Populations' and review earlier questions that can now be partly answered.	Say: 'Which questions on our board can we now start to answer? What new questions do your counts raise?' Give WAIT TIME of 10-15 seconds. Cold-call 3 groups to read a sticky note aloud and ask where it fits. Highlight the question that links to the next lesson: 'Is the distribution of grass uniform throughout the school field?' Ask: 'Did our two 10 m x 10 m areas tell us about the whole field?' Move one answered question to the 'What we have figured out' column.	Driving Question Board: learners see their understanding grow and take ownership of the next steps of the inquiry. Sorting questions as answered, partly answered, or new helps them connect each lesson's evidence to the overall phenomenon of disappearing fish.	Check that sticky notes (1) use population or density vocabulary correctly, (2) pose testable questions, and (3) connect to the river phenomenon. Note which groups ask questions about distribution or sampling, a sign they are ready for the next lesson.

Model Building Phase  VIDEO: Predator-prey cycles <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Population ecology <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(5 min) Learners revise their class model of the river ecosystem. In notebooks, they add a new element: 'Population size and density can be measured by counting individuals in a known area.' They draw the river with fish, algae, and weeds, and add a labelled box showing how scientists might count or estimate fish and algae to check whether numbers are changing. They then complete an exit ticket individually: (1) Define population density. (2) Write the density for a site with 180 plants in 100 m². (3) Give one situation where direct counting is suitable and one where it is not. (4) One question you still have.	Say: 'Add one new idea from today to your model and show how it helps explain what is happening in the river.' Give WAIT TIME of 10-15 seconds before learners begin drawing. Walk around and ask 3 learners: 'Where in your model would you place the counting?' Distribute exit tickets and collect them at the door. Close by stating the new question: 'Our counts came from two small areas. Is the distribution of grass uniform throughout the school field? We will find out next lesson.'	Cumulative model revision with an exit ticket: adding population measurement to the river model helps learners link a classroom method to a real ecological problem. The exit ticket allows each learner to show individual understanding after group work.	Exit ticket (as requested for Lessons 3 to 6). Success indicators: (1) defines population density as the number of individuals per unit area; (2) correctly calculates 1.8 per m² for 180 plants in 100 m²; (3) gives a suitable and an unsuitable example of direct counting (for example, counting trees in a small plot is suitable, counting fish in Lake Victoria is not); (4) writes a relevant question. Sort the tickets into three piles: secure, developing, and needs support, and use them to plan the start of Lesson 4.
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D. TEACHER REFLECTION

After the lesson, consider: Did every learner take an active role in the jigsaw, or did some depend on others? Were the counts accurate, and did groups agree on what counted as one individual? Which learners confused population size with density? Did learners link the school counts to the river with fish, algae, and weeds? Check how many exit tickets show correct density calculations and identify the learners who need help with area units. Decide whether to give extra time to marking areas or to division. Note which questions on the DQB point toward distribution and sampling, and use them to open Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We marked two 10 m x 10 m areas in the school compound, counted every individual of one organism (such as grass tufts or weeds), and recorded the number, area and density for each site. The sites differed in population size and density.
What did I learn?	A population is all the individuals of one species in an area. Population density is the number of individuals per unit area (number / area). Direct counting is suitable when the population is small and individuals are easy to identify and count.
How does this explain the phenomenon?	Sites differed because of factors such as sunlight, soil, water and trampling. Direct counting works for the school compound but not for fish hidden in a river, so other methods are needed. We now ask: Is the distribution of grass uniform throughout the school field?

LESSON 4 (80 minutes (2 x 40-minute periods)): Quadrat Sampling: Is the Distribution of Grass and Weeds Uniform Across the School Field?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use 1 m x 1 m quadrats placed randomly to estimate the population size and density of grass and weeds on the school field, and use this evidence to explain how plant populations can differ with environmental conditions, linking to the booming algae and aquatic weeds in the Kenyan river.
Knowledge	By the end of the lesson, the learner should be able to: (d) estimate population sizes of organisms in ecosystems using the quadrat sampling method; (c) describe factors affecting population sizes in ecosystems.
Skills	Planning and carrying out investigations (random sampling, counting, recording); analysing and interpreting data (calculating mean number per quadrat, density, and total population estimate); using mathematics (area, mean, density); collaborating in jigsaw groups.
Attitudes	Integrity: utilises resources prudently while sampling and does not trample or uproot the school field unnecessarily. Citizenship: appreciates that careful data helps the community solve local problems such as river degradation. Patriotism: shows care for Kenyan ecosystems.
Key Inquiry Question	How do organisms interact with their environment?
Purpose in Storyline	Lessons 1-3 established that organisms are adapted to habitats and that populations are shaped by resources. Lesson 4 gives the first quantitative method of estimating population size and density. Learners see that a population is not spread evenly, which prepares them to explain why algae and weeds boom in some parts of the river and fish decline in others. It ends with the new question: How does the population of plants change as environmental conditions change across the school compound?
Safety Notes	Work only in the agreed area of the school field and stay in sight of the teacher. Wear closed shoes and avoid areas with broken glass, thorns or stagnant puddles. Wash hands with soap after handling soil and plants. Do not uproot or taste any plant. Check for learners with allergies to pollen or grass. Return all equipment. Avoid sampling in the hottest part of the day and keep drinking water available.

B. LESSON OVERVIEW

Learners recall the Kenyan river video (2:20-4:50) where fish are declining while algae and aquatic weeds boom, and predict whether plants are spread evenly in a habitat. In jigsaw groups they use 1 m x 1 m quadrats at random locations on the school field, count and record grass and weed plants, then calculate mean number per quadrat, density and estimated total population. Expert groups compare results from sunny, shaded, wet, compacted and trampled spots, then share with home groups. The Driving Question Board is updated with questions about distribution and conditions, and the class revises its model of how populations respond to the environment. An exit ticket checks understanding and leads to Lesson 5.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners recall the river video	Display a 10-second clip or	Prediction with reasoning (Think-	Observable indicators: learner

 VIDEO: 01 Biotic and Abiotic Factors Source: Seavuria  Search ARES for similar videos  HTML: Population Size, Density, and Distribution (Flexbook) Source: CK-12  Search ARES for similar readings	(2:20-4:50), where fish are fewer but algae and weeds are thick near the banks and the water smells. Each learner writes a prediction in their notebook: 'Is the distribution of grass and weeds uniform throughout the school field? Give one reason.' They sketch a top view of the field showing where they think plants are dense or sparse. Pairs share, then groups of 4 agree on a prediction. They also predict: 'If weeds grow thickly in only part of a river, what does that suggest about conditions there?'	screenshot from the river video and say: 'Remember the green weeds crowding the river edge. Are they spread evenly along the whole river?' Give WAIT TIME of 10-15 seconds before anyone speaks. Say: 'Write your prediction silently first.' Cold-call 4 learners from different groups: 'What did you predict, and what is your reason?' Record predictions on the board without judging, such as 'even everywhere', 'more where there is water', 'more in shade'. Ask: 'How could we count plants on a whole field without counting every single one?' Hold the quadrat up and say: 'This is our tool. Today it is our evidence.'	Pair-Share): Making predictions public exposes preconceptions (for example that plants are evenly spread), creating a need for data. Learners revisit these predictions at the end to see how evidence changed their thinking.	predictions include a reason referring to a factor such as light, water, soil or trampling; at least 3 of 4 cold-called learners link their prediction to the river phenomenon; learners suggest sampling rather than counting every plant. Teacher notes on a class list those who hold the misconception that plants are evenly distributed.
Observe Phase  VIDEO: Techniques for random sampling and avoiding bias Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Population Size, Density, and Distribution Practice Source: CK-12  Search ARES for similar readings	Jigsaw Step 1 (expert groups): the class is divided into five expert teams, each assigned a zone of the school field, for example (A) open sunny area, (B) under a tree or shade near a building, (C) near a tap, drain or damp area, (D) a footpath or football pitch area with heavy trampling, (E) edge near the fence or compost area. Each team uses a 1 m x 1 m quadrat (made from sticks and string or a metre-rule frame). They place the quadrat at 5 random locations in their zone by tossing a bottle top or using random numbers for steps along two sides. For each quadrat they record: (i) number of grass tufts or individual weed plants (for example Kikuyu grass, black jack/Bidens pilosa, Sodom apple, goat weed), (ii) the species, (iii) the percentage cover estimate, and (iv) notes on abiotic conditions such as light, soil moisture (touch test) and trampling. They then calculate: total counted, mean per quadrat, density = mean number	Before going out, model the technique with 'Quadrat sampling means placing a square frame of known area at random places so we count a small part and estimate the whole.' Ask: 'Why must we place the quadrat randomly, not where the grass looks nicest?' WAIT TIME 10-15 seconds, then cold-call 3 learners. Demonstrate random placement by tossing an object backwards over the shoulder. State safety rules and the agreed boundaries. Circulate and prompt: 'How many plants are inside the frame? Do you count a plant on the line?' (agree a rule: count plants touching the top and left edges only). 'What does the soil feel like here compared with the other zone?' Check that each team has at least 5 quadrats and that roles are assigned (placer, counter, recorder, timekeeper). Back in class, collect each team's means on the board table: Zone Mean per quadrat Density per m2 Estimated population.	Jigsaw (expert groups): Each expert group becomes responsible for evidence from one zone, so every learner holds unique data needed by the whole class. This builds interdependence and mirrors how Kenyan ecologists (for example researchers at Kenya Wildlife Service or the National Museums of Kenya) sample sections of a habitat and combine results.	Observable indicators: quadrats are placed randomly (not at chosen spots); counts are recorded in an organised table; the counting rule at quadrat edges is applied consistently; mean and density calculations are correct (density = total count / (number of quadrats x 1 m²)); learners record at least one abiotic observation per zone. Teacher uses a checklist while circulating.

	per m ² , and estimated population = density x total area of the zone (for example 200 m ²). They record results in a class data table.			
Explain Phase  VIDEO: 01 Biotic And Abiotic Factors Source: Seavuria  Search ARES for similar videos  HTML: Population Size, Density, and Distribution (Flexbook) Source: CK-12  Search ARES for similar readings	Jigsaw Step 2 (home groups): learners regroup so each home group of 5 has one expert from every zone. Each expert presents their zone's data and abiotic observations. The home group completes a comparison table and answers: (1) Is the distribution of grass and weeds uniform across the field? (2) Which zones have the highest and lowest density and what factors might explain this? (3) How does the quadrat method give an estimate of population size and why is it only an estimate? (4) How could the same method help the community study weeds and algae in the river, and why might weeds thrive in some parts of the river (for example near where fertiliser or sewage from farms in Kisumu or Nakuru enters) but fish decline? Groups write a claim-evidence-reasoning statement on a chart paper, for example 'Plant density differs between zones because ... (evidence: mean of 42 per m² in the damp zone versus 6 in the trampled zone).'	Say: 'Experts, you are now the teachers. Share your numbers first, then your conditions.' Circulate and ask: 'What is your evidence for that claim?' 'Could chance explain the difference? How could you reduce the effect of chance?' (more quadrats). Prompt: 'Connect this to our river. If weeds crowd one stretch of the river, which conditions might be different there?' WAIT TIME 10-15 seconds, then cold-call 4 groups to share their claim. Introduce terms verbatim from the content: 'population', 'habitat', 'carrying capacity', 'biotic and abiotic factors'. Clarify that the correlation seen here does not prove the cause; further tests are needed (links to Lesson 5). Correct misconceptions such as 'more plants always means a healthier ecosystem' by noting that a boom in algae and aquatic weeds in the river may signal nutrient enrichment that harms fish by reducing oxygen.	Jigsaw (home groups) with Claim-Evidence-Reasoning: Combining zone data forces learners to compare, find patterns, and connect measured differences to environmental factors. CER structures the argument and links field data to the river phenomenon.	Observable indicators: groups state clearly that distribution is not uniform and cite numerical evidence; they name at least two factors (for example light, water, soil nutrients, trampling, competition) that affect plant populations; they explain that sampling gives an estimate; they make at least one link between the field patterns and algae or weed growth in the river. Teacher listens for correct use of the words density, population and random.
Driving Question Board (DQB) Creation  VIDEO: Techniques for random sampling and avoiding bias Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners return to the Driving Question Board: 'Why are fish disappearing from a water body that once supported many fish, and what can the community do to restore the ecosystem?' Individually they write one new question on a sticky note arising from today's data, for example 'Why were there more weeds near the tap?', 'Do weeds in the river also grow more where there is runoff?', 'How would we sample	Say: 'Take two minutes to write a question that today's data made you curious about.' WAIT TIME 10-15 seconds after reading the driving question aloud. Invite 3 learners to read their notes and explain where they would place them. Cold-call 2 more learners to identify an earlier question that can now be partly answered, for example 'Are the weeds evenly distributed?'. Highlight and star the question: 'How does the population	Driving Question Board: Learners track growing understanding and see that their questions drive the next investigation. Sorting questions into categories helps them recognise how population data connects to environmental factors and human activity.	Observable indicators: sticky-note questions are linked to evidence from the quadrat data; at least half of the class contribute a question that goes beyond recall (for example a cause-and-effect question); learners can correctly categorise questions; some questions mention human activities such as fertiliser runoff, sewage, or washing near the river.

 EXERCISE: Population Size, Density, and Distribution Practice Source: CK-12  Search ARES for similar readings	plants in moving water?' They place notes on the DQB under categories (Population size, Environmental factors, Human activity, Restoration). They also mark questions that have been partly answered with a tick.	of plants change as environmental conditions change across the school compound?' and say: 'This is our question for Lesson 5.' Connect: 'Our school field is a small model of the river. What patterns might also exist along the river?'		
Model Building Phase  VIDEO: Ecology introduction <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Stratified Random Sampling Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Groups revise the class model of the river ecosystem started in earlier lessons. They add a section on plants: draw the river with banks, fish, algae, aquatic weeds, and labelled arrows showing factors (light, nutrients, oxygen, water flow, human activity). They add a new component: 'Plant populations are not evenly distributed; density varies with environmental conditions and can be estimated by sampling.' They add a note on how quadrat sampling (or its adaptation to a section of the riverbank) could estimate the weed population. They label unknowns with question marks, for example 'Why do nutrients cause algae to boom?' Each learner then completes the exit ticket individually.	Say: 'Look at our model. What can we now add about plants and where they grow?' WAIT TIME 10-15 seconds, then cold-call 3 learners to suggest additions. Prompt: 'Which arrow in our model do your quadrat data support?' 'What part of the model are we still unsure about?' Ask groups to use a different colour for new additions. Distribute the exit ticket with 3 prompts: (1) Define population density and show how you calculate it. (2) Give two reasons why plant density may differ between two parts of a habitat. (3) Explain how what you learned today helps to explain the weeds and algae in the Kenyan river. Collect tickets at the door and sort them into 'secure', 'partly secure' and 'not yet' piles to plan Lesson 5.	Cumulative model revision with jigsaw sharing: Learners integrate new evidence into a shared model, which makes the relationship between organisms and their environment visible. Marking uncertainties keeps the model open to testing in later lessons.	Exit ticket. Observable indicators: correct statement of density as number of individuals per unit area (for example plants per m²) with a worked calculation; at least two valid factors named (light, water, soil nutrients, trampling, competition); a reasonable link to algae or weeds in the river; mention that quadrat results are estimates. Teacher records the percentage of learners who meet each indicator and plans reteaching of random sampling or calculations if fewer than 70% succeed.

D. TEACHER REFLECTION

Did all groups place quadrats truly randomly, or did some choose convenient spots? Which learners struggled with the density and population estimate calculations, and what scaffolds (worked example, calculator pairing) helped? Did the zone comparison produce clear differences, or should I choose more contrasting zones next time? How well did learners connect the school field to the river phenomenon, and who needs more support linking abiotic factors with plant growth? Were the jigsaw roles shared equitably, including quieter learners? Use the exit ticket results to adjust the Lesson 5 investigation on environmental conditions.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We used 1 m x 1 m quadrats placed randomly in different zones of the school field and counted grass and weed plants. We saw
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	that the numbers per quadrat were very different in sunny, shaded, damp and trampled areas.
What did I learn?	Plants are not spread evenly in a habitat. Population density is the number of individuals per unit area, and quadrat sampling lets us estimate population size without counting every plant. Light, water, soil and trampling affect the size of plant populations.
How does this explain the phenomenon?	Just as weeds were dense in some parts of the school field because of local conditions, algae and aquatic weeds boom in parts of the river where conditions, such as nutrients from human activity, favour them. This helps explain why the river is full of life but fish are disappearing, and leads us to ask how plant populations change as environmental conditions change.

LESSON 5 (80 minutes (double lesson)): Belt Transect Across a Changing Habitat: How Does Plant Population Change as Conditions Change?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use a belt transect across the school compound (open field, shaded area, pathway, garden) to estimate and compare plant abundance, and relate differences in population size to abiotic and biotic factors. They then apply this thinking to the disappearing fish and the booming algae and weeds in the river.
Knowledge	By the end of the lesson, the learner should be able to: (c) describe factors affecting population sizes in ecosystems; (d) estimate population sizes of organisms in ecosystems using sampling methods (belt transect, quadrat). Learners explain that population size varies across a habitat because abiotic factors (light, soil moisture, soil nutrients, trampling) and biotic factors (competition, grazing, human activity) differ from place to place.
Skills	Planning and carrying out an investigation (laying out a belt transect with two parallel lines and quadrats at regular intervals); collecting and recording data in a table; calculating mean abundance and density; analysing and interpreting data (graphing abundance against position); constructing explanations; communicating clearly with peers in expert and home groups.
Attitudes	Core competencies: Self-efficacy (communicating clearly with others while discussing findings) and Citizenship (solving problems in the local community). Values: Integrity (using resources prudently, returning the site to its original state, recording honestly). PCI: Prevention of Global Warming and Waste Management (links to how human activity changes habitats). Learners show curiosity, respect for living things, and care for the school environment.
Key Inquiry Question	How do organisms interact with their environment? What factors affect population size in an ecosystem?
Purpose in Storyline	Lessons 1-4 established that organisms are adapted to habitats and that ecological terms such as population, habitat and niche describe how organisms live. In Lesson 5, learners gather first-hand quantitative evidence that population size changes across a habitat because environmental conditions change. This gives them a method (sampling) and a concept (limiting factors) that they will use to explain why fish numbers fall while algae and weeds increase in the river. It ends with the question: What factors affect population size in an ecosystem? and leads into Lesson 6.
Safety Notes	Take learners out in groups of 5-6 with clear boundaries. Wear closed shoes and long sleeves where possible. Check the area beforehand for snakes, thorny plants, stinging nettles, broken glass, ant nests and sharp objects. Learners with allergies to pollen or plants should handle only the recording sheets. Do not uproot or pick plants; count them in place. Wash hands after the fieldwork. Carry water on hot days and avoid doing the activity at midday. Stay within sight of the teacher and use the agreed whistle or call to return. Pathways used by vehicles should be avoided.

B. LESSON OVERVIEW

The lesson begins by revisiting the river video (2:20-4:50), where fish are fewer while algae and weeds boom. Learners predict how plant populations will differ across the school compound (open field, shaded area, pathway, garden) and why. Working in jigsaw groups, learners lay out a belt transect using two parallel string lines, place quadrats at regular intervals, and each expert team measures one factor or one measure (plant counts, light, soil moisture, trampling and soil observation). Expert members then return to home groups to combine the data, graph plant abundance across the transect and explain the pattern. Learners link this to the river: the river is also a habitat with changing conditions (nutrients from farms and sewage, light, oxygen) that favour algae and weeds but not fish. The class adds new questions to the Driving Question Board and revises the cumulative model. An exit ticket checks understanding of what affects population size.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 01 Biotic And Abiotic Factors Source: Seavuria  Search ARES for similar videos  HTML: Population Size, Density, and Distribution (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the river video (2:20-4:50): fewer fish, more algae and aquatic weeds, and a bad smell. The teacher poses the school-compound question: 'How does the population of plants change as environmental conditions change across the school compound, from open field to shaded area to pathway to garden?' Individually, learners sketch a quick line graph or bar prediction of plant abundance across the four zones and write one sentence explaining why. They then share predictions with a partner and note one agreement and one disagreement. They also write a quick guess about how this relates to the river.	Say: 'Remember the river near our community, where fish are fewer but algae and weeds are everywhere. Today we will study a place we can touch: our school compound. Before we go out, I want your best prediction.' Show a simple sketch on the board of the four zones in a row. Ask: 'Where do you think we will find the most plants per square metre, and where the fewest? Why?' Give WAIT TIME of 10-15 seconds and then allow 2 minutes of silent writing. Cold-call 4 learners to share predictions, asking each: 'What condition in that zone is making you say that?' Record the conditions they mention (light, water, trampling, soil) in two columns labelled Abiotic and Biotic. Do not correct predictions; say: 'We will test these ideas with data.'	Prediction with reasoning (Predict-Observe-Explain). Learners commit to a prediction and a reason before collecting data, which surfaces their prior ideas about limiting factors and makes the later evidence more meaningful when it confirms or challenges them.	Look for: learners' predictions that name at least one specific factor (for example, 'shade means less sunlight, so fewer grasses' or 'the pathway is trampled'); use of correct terms such as abiotic and biotic; and predictions that differ between zones rather than being the same everywhere. Note learners who say 'plants need water' without linking it to a zone and prompt them with 'Which zone would be wettest or driest, and why?'. Record on a class list the number of learners who connect the compound to the river.
Observe Phase  VIDEO: 01 Biotic And Abiotic Factors Source: Seavuria  Search ARES for similar videos  EXERCISE: Population Size, Density, and Distribution Practice Source: CK-12  Search ARES for similar readings	Learners go outside in groups of 5-6 to the transect area. As a class, they stretch two parallel lines (for example, 10 m long and 1 m apart) across the four zones, and place a 1 m x 1 m quadrat (or a 50 cm x 50 cm quadrat made from sticks or wire) at regular intervals (every 2 m) between the lines. Jigsaw expert roles: Team A counts and records plant abundance (number of individual plants, or percentage cover, for example of Kikuyu grass, Sodom apple, black jack, sukuma wiki or maize in the garden) in each quadrat; Team B measures light (using a simple	Before going out, model how to place a quadrat: 'Place it flat on the ground at the mark, do not throw it, and record what is inside the frame only.' Check roles with: 'Team B, what exactly will you measure, and in what units?' Walk between groups, asking: 'Why do we place quadrats at regular intervals rather than choosing the greenest spots?' (to avoid bias) and 'What will happen to your results if two people count differently?' Allow WAIT TIME of 10-15 seconds after each question. Cold-call 3 groups to explain how they decided which	Jigsaw. Each learner becomes the only expert on one set of data (plants, light, soil moisture or trampling) and must teach it to the home group. This makes everyone accountable for part of the evidence and builds a complete picture only when all data are combined, just as understanding the river requires combining evidence on nutrients, light, oxygen and fishing.	Observe whether learners place quadrats at the marked intervals without bias, count consistently and record data in the correct column with units. Check that every learner in the expert team can explain how their measurement was taken. Listen for learners comparing zones (for example, 'the garden quadrats have more weeds than the pathway'). Use a checklist: correct quadrat use (yes/no), data table complete (yes/no), every member records (yes/no). Note groups that need support with counting rules such as how to count plants on the

	shade-or-sun rating, or a lux app if available); Team C measures soil moisture (feel test or a simple soil squeeze scale 1-5, or the mass of soil before and after drying if possible) and soil colour or humus; Team D records trampling and human activity (footprints, bare ground, litter, tools, signs of grazing). Each expert team records data in a table by position. Learners take care to count each plant the same way and to place quadrats without bias. They then return indoors and form home groups with one expert from each team.	plants to count. Remind groups: 'Do not uproot plants; leave the area as you found it.' Use a whistle to call groups back. In the classroom, give the jigsaw instructions: 'Each expert teaches your home group your data in 3 minutes. Others write it in the master table.' Provide a master data table with columns: Quadrat position, Zone, Plant count or cover, Light, Soil moisture, Trampling/human activity.		quadrat edge.
Explain Phase  VIDEO: Ecology introduction <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Population Size, Density, and Distribution (Flexbook) Source: CK-12  Search ARES for similar readings	In home groups, learners combine data in the master table, calculate the mean abundance for each zone (total plants divided by number of quadrats) and estimate density (plants per square metre). They draw a bar graph or line graph of plant abundance against position along the transect and mark the zones (open field, shaded area, pathway, garden). Using the data on light, moisture and trampling, they write a claim-evidence-reasoning explanation: 'Plant population is highest/lowest in ____ because ____.' They compare the result to their earlier prediction. Then they use the river: 'What conditions in the river (extra nutrients from farm fertiliser, sewage, soap, and warm sunlit water) could favour algae and weeds, and what conditions could reduce fish?' Groups produce a short comparison table: school compound factors vs river factors, and note that human activity (trampling, gardening, farming runoff) changes the population size of different organisms.	Say: 'Experts, you now have three minutes each. Home group, your job is to find the pattern.' Ask: 'Which zone had the highest plant density? Which had the lowest? What else was different in that zone?' Give WAIT TIME of 10-15 seconds and then cold-call 5 learners from different groups, each giving one piece of evidence. Probe: 'Is that correlation or cause? How could we test it?' and 'Why might the pathway have fewer plants even though it gets lots of light?' (trampling, compaction). Then connect: 'In the garden, someone adds manure and water. What does this remind you of in the river?' Guide learners toward ideas of nutrient enrichment causing algae and weeds to boom, and limiting factors (light, nutrients, space, trampling) controlling population size. Introduce the terms limiting factor, population density and abundance. Address misconceptions such as 'more plants always means a healthier habitat' by asking: 'Are all the	Claim-Evidence-Reasoning with data analysis, supported by jigsaw home-group discussion. Learners must justify claims with their own transect data, then transfer the reasoning pattern (conditions change, so population size changes) to the river phenomenon. This builds the central idea that population size depends on a combination of abiotic and biotic factors.	Exit ticket (as planned for Lessons 3-6): On a half sheet of paper, learners answer: (1) Name two factors that affected plant population size across our transect and give the evidence. (2) Give one biotic and one abiotic factor that could affect fish or algae populations in the river. (3) One question you still have. Success indicators: correct use of data (for example, 'mean of 4 plants/m² in the shade compared with 18 plants/m² in the open field'), correct classification of factors, and a link to the river. Sort the slips into three piles: secure, partial and needs support; use the piles to plan the start of Lesson 6.

		plants in the garden useful? Which are weeds?' Close with: 'Which factors are abiotic and which are biotic?'		
Driving Question Board (DQB) Creation  VIDEO: 6a-c. Population ecology <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Population Size, Density, and Distribution Practice Source: CK-12  Search ARES for similar readings	Learners return to the Driving Question Board with the question 'Why are fish disappearing from a water body that once supported many fish, and what can the community do to restore the ecosystem?' Each group writes one new question on a sticky note based on the transect, for example: 'Does extra fertiliser increase algae the way manure increases weeds in the garden?', 'Why does a trampled pathway have few plants, and what is the river's version of trampling?', or 'How can we count fish in a river?' They place questions under categories such as Factors affecting population, Sampling methods, Human activity and Solutions. They also move any earlier questions that have been partly answered into an 'Answered so far' column and add the evidence.	Say: 'Look at the DQB. What did our transect help us understand, and what new questions has it raised?' Give WAIT TIME of 10-15 seconds, then ask each group to place one sticky note. Cold-call 3 groups to read their questions aloud and say which category they belong in. Ask: 'Which earlier question can we partly answer now? What is the evidence?' Highlight questions about nutrient enrichment, sampling in water, and community action for use in later lessons. Praise good questions such as 'How would we do a transect in a river?' and say, 'That is exactly the kind of question scientists ask.'	Question clustering on the Driving Question Board. Sorting and linking new questions to the anchoring phenomenon makes learners' growing understanding visible and keeps the investigation purposeful and learner-driven.	Check that questions are testable and linked to the river phenomenon (not simply 'What is a quadrat?'). Count how many groups moved or answered at least one earlier question using evidence. Look for learners using terms such as population, limiting factor, abiotic and biotic correctly when presenting their notes. Note learners who need help turning a statement into a question.
Model Building Phase  VIDEO: 01 Biotic and Abiotic Factors Source: Seavurria  Search ARES for similar videos  EXERCISE: Population Size Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES	Learners revise their cumulative class or group model of the river ecosystem. They add a new section on factors affecting populations: arrows showing how light, nutrients, oxygen, space, competition and human activities (farm runoff, sewage, washing, deforestation along the banks, overfishing) affect algae, aquatic weeds and fish populations. They add labelled boxes such as 'More nutrients leads to more algae and weeds' and use dotted arrows for ideas that are still uncertain (for example, why the smell appears). They add a key linking the transect	Say: 'Take out your model of the river. Add what you have learned about what controls population size, using evidence from our transect.' Circulate and ask: 'Which factor in your model is like the shade on our transect, limiting growth? Which is like the manure in the garden, boosting it?' Allow WAIT TIME of 10-15 seconds. Cold-call 3 groups to explain one new arrow, asking: 'What evidence supports that arrow?' Encourage learners to use different colours for abiotic and biotic factors. Finish with: 'Our model still cannot fully explain why the fish are	Cumulative model revision. Learners integrate new evidence into an evolving model of the river, showing relationships between factors and populations. This makes causal reasoning explicit and shows gaps to be investigated in later lessons.	Look for models that include at least two abiotic and two biotic factors, arrows that show direction of influence, and at least one link to human activity. Check that learners label the evidence source (transect data) and mark uncertain relationships. Use a quick rubric: 3 = factors, arrows and evidence all included; 2 = factors and arrows only; 1 = factors listed without relationships. Note learners who show cause and effect incorrectly, such as 'fish eat algae therefore algae cause fish to disappear', and address this in Lesson 6.

for similar readings	to the model: 'Different conditions lead to different population sizes.' They then write one sentence under the model: 'What I can now explain / What I still need to find out.'	disappearing. What do we need to know next?' and link to the next lesson with: 'What factors affect population size in an ecosystem, and how can we estimate numbers of animals like fish that we cannot easily see?'		
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D. TEACHER REFLECTION

After the lesson, consider: Did learners place quadrats and count plants consistently? Which groups connected the pattern across the transect to specific abiotic or biotic factors, and which only described the pattern? Were learners able to transfer the idea of limiting factors from the school compound to the river? Which exit ticket responses showed misconceptions, such as thinking that more plants always means a better habitat, or that only one factor controls a population? How well did the jigsaw work: did every expert contribute and could every learner explain the combined data? What adjustments are needed for time, quadrat size or measurement of soil moisture and light? Plan Lesson 6 around the gaps you identified.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We stretched two parallel lines across the school compound (open field, shaded area, pathway, garden), placed quadrats at regular intervals and recorded plant abundance, light, soil moisture and trampling. Plant numbers differed from zone to zone (record the actual mean per square metre for each zone).
What did I learn?	Population size changes across a habitat because conditions change. Abiotic factors (light, water, soil nutrients) and biotic factors (competition, grazing, human activity such as trampling and gardening) affect how many individuals can live in a place. A belt transect with quadrats is a sampling method used to estimate abundance.
How does this explain the phenomenon?	Like the plants in the compound, populations in the river respond to changing conditions. Extra nutrients from farms and sewage can favour algae and weeds, while changes in oxygen, light and human activity can reduce fish. This helps explain why the river is still full of life but fish are disappearing. We still need to find out which factors matter most and how to estimate fish populations.

LESSON 6 (80 minutes (2 x 40-minute periods)): How Many Fish Are Left? Capture-Recapture Simulation to Estimate Population Size

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use a capture-recapture simulation with 100 counters to estimate the size of a population they cannot count directly, and apply the method to judge whether fish numbers in the Kenyan river have fallen.
Knowledge	Learners estimate population sizes of organisms in ecosystems (KICD SLO d). Estimated population = (number initially marked x number in second sample) / number of marked individuals recaptured. Direct counting is impractical for mobile organisms such as fish in a river or Lake Victoria.
Skills	Using models and simulations (NGSS SEP: Developing and Using Models); using mathematical thinking to calculate and compare estimates; planning and carrying out investigations with repeated trials; analysing and interpreting data; communicating clearly in groups.
Attitudes	Core competency: Self-efficacy - the learner communicates clearly with others in the group. Value: Integrity - the learner uses resources (counters, containers) prudently and records data honestly. Citizenship: the learner uses evidence to help the community make decisions about river fish stocks. PCI: Waste Management and Environmental Education - polluted water reduces fish populations.
Key Inquiry Question	How do organisms interact with their environment?
Purpose in Storyline	In Lessons 1-5 learners met the river phenomenon (fewer fish, more algae and weeds, smell) and learned about habitats, adaptations and ecosystem features. The class now asks: 'How would we know how many fish are in the river, and whether the number is falling?' This lesson gives them a tool to turn the claim 'fish are disappearing' into numbers, and it raises the next question: 'What factors affect population size in an ecosystem?'
Safety Notes	Use small counters, beans or bottle tops only; ensure learners do not put them in mouths. Keep floors clear of dropped counters to prevent slips. Containers should be stable on desks. If the practical is extended outdoors to a stream or pond, do not go near water without teacher supervision.

B. LESSON OVERVIEW

Learners watch again the river clip (2:20-4:50) and predict how they could find out how many fish remain when fish cannot be counted one by one. In expert-then-home jigsaw groups they run a capture-recapture simulation: 100 counters (the 'fish') are in a container, 20 are captured and marked (replaced with a different colour), mixed back, and a second sample of 20 is taken. From the marked ones recaptured (for example 5), they calculate the estimate = $(20 \times 20) / 5 = 80$ and compare it with the true value of 100. They repeat several trials, pool class data, discuss accuracy and assumptions (thorough mixing, no deaths or migration, marks not lost), and relate the method to estimating fish in the river. Learners complete an exit ticket, add questions to the Driving Question Board and revise their class model, leading to the next question: what factors affect population size in an ecosystem?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Worked example: Logistic model word problem <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Population Size, Density, and Distribution (Flexbook) Source: CK-12  Search ARES for similar readings	Learners replay the river video segment (2:20-4:50) and recall that fishers say the catch is smaller. Individually they write a prediction in their notebooks: 'How could we find out how many fish are in the river if we cannot catch and count every one? Would our answer be exactly right?' They share with a partner, then a few ideas are shared with the class (e.g. count the ones caught in a net, multiply, mark some fish). Learners also predict: 'If we mark 20 out of 100 and later 5 of a catch of 20 are marked, what does that tell us about the whole population?'	Say: 'Remember the river in the video. Fishers say there are fewer fish. But how can anyone prove it, when the fish are hidden under the water?' Play 2:20-4:50 again with no commentary. Give WAIT TIME of 10-15 seconds after asking: 'Why can't we just count all the fish?' Direct learners to write silently for 2 minutes, then think-pair-share for 2 minutes. Cold-call 4 pairs: 'Which method did you suggest, and why?' Record all predictions on the board without judging them. Ask: 'What would be our reason for trusting a number?'	Predict-Share-Revisit: Making predictions publicly commits learners to ideas that they can test later, and surfaces the preconception that populations must be counted one by one. Recording ideas lets learners compare predictions with simulation results, building understanding of why sampling is needed.	Look for: (1) learners write a prediction with a reason rather than 'I don't know'; (2) at least 3 of 4 cold-called pairs name an approach (sampling, marking, counting a part); (3) some learners recognise that a count of the whole river is impractical. Note learners who think the number must be exact; revisit them in the Explain Phase.
Observe Phase  VIDEO: 6a-c. Population ecology <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Population Size, Density, and Distribution Practice Source: CK-12  Search ARES for similar readings	Learners form groups of 4 (home groups) and each learner takes one jigsaw role as an expert: Capturer (takes the first sample), Marker (swaps the captured counters for a different colour or marks them), Recapturer (mixes and takes the second sample), Recorder/Calculator (records and computes). Procedure: (1) Put 100 counters (e.g. beans or bottle tops of one colour) in a container. (2) Without looking, randomly take 20 counters. (3) Mark/replace them with a different type or colour. (4) Return and mix thoroughly. (5) Take a second random sample of 20. (6) Count how many marked ones are recaptured. Example: marked = 20, second sample = 20, recaptured = 5. Estimated population = $(20 \times 20) / 5 = 80$. Groups repeat for 5 trials (returning all counters each time) and record data in a table: Trial Marked Second sample	Before starting, show the set-up and model one trial in front of the class. Say: 'Pretend the counters are tilapia in the river. Marking them is like tagging fish with a small coloured tag from the fisheries officers.' Circulate with a checklist. Ask groups: 'Did you mix thoroughly before the second sample? What happens if you do not?' and 'What do you notice about the estimates across trials?' Give WAIT TIME of 10-15 seconds after each question. Cold-call 3 groups to report the number recaptured in trial 1 and their estimate. After the jigsaw exchange, ask: 'What differences did you notice in other groups' results?' Insist on honest recording of results, even if the estimate is far from 100. Ensure equipment is returned and the area tidy.	Jigsaw groups: Each learner has a role and becomes an expert in one step, and then compares data with learners from other groups. Teaching others the step and hearing different results makes the variation in estimates visible and builds a shared understanding of how sampling works.	Check that: (1) learners correctly apply the formula in at least 4 of 5 trials; (2) the table is complete with labelled columns; (3) groups record different estimates for different trials and mention that results vary; (4) learners mix counters before the second sample. Ask 2 learners per group to explain the meaning of 'recaptured' in their own words.

	Recaptured Estimate. They find the mean estimate and compare it with the true value of 100. Groups swap roles between trials, and then one member from each group moves to another group (jigsaw) to share their results and method differences.			
Explain Phase  VIDEO: Worked example: Logistic model word problem <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Population Size Practice Source: CK-12 Search ARES for similar readings	Groups pool the class estimates on the board (e.g. 80, 95, 110, 100, 125) and calculate the class mean. They discuss: Why did estimates differ from 100? Why does the estimate become better with more trials or larger samples? What happens to the estimate if more marked individuals are recaptured (population smaller) or fewer (population larger)? Learners then apply the method to a Kenyan fish scenario: 'Fisheries officers on Lake Victoria catch and tag 50 Nile tilapia, release them, and later catch 60 fish, of which 6 are tagged. Estimate the population.' ($50 \times 60 / 6 = 500$). They compare with a second year in which 50 tagged, 60 caught, 12 tagged (estimate 250) and explain what this suggests about fish numbers. They list the assumptions of the method (marked individuals mix fully, no births, deaths or migration between samples, marks are not lost, marking does not harm fish) and explain how pollution or overfishing might break them. They link to the river phenomenon: 'How could this method show whether fish are truly disappearing?'	Draw the class results on the board and ask: 'What do you notice about these estimates?' Give WAIT TIME of 10-15 seconds. Ask: 'Why is the average of many trials more reliable than one trial?' Cold-call 5 learners for different ideas. Pose the Lake Victoria problem and have learners calculate in groups, then ask 3 groups to show the working. Challenge: 'If the second-year estimate is 250 and the first was 500, what can we say about the fish? What else must we check before we blame fishing, pollution, or both?' Probe with: 'What does this method NOT tell us?' (It does not tell us why the numbers dropped.) Correct the misconception that the estimate is an exact count. Ask learners to explain, using the formula, why more recaptured marked individuals means a smaller population.	Claim-Evidence-Reasoning with data pooling: Learners combine class data to spot patterns, then justify claims about population change using calculations and stated assumptions. Applying the formula to a Kenyan lake scenario transfers the model to the real phenomenon and shows both the power and the limits of sampling.	Observable indicators: (1) learners correctly calculate 500 and 250 for the lake scenarios; (2) they state that a smaller proportion of tagged fish in the sample means a larger population, and vice versa; (3) they name at least two assumptions; (4) they say that the method estimates numbers but does not explain the cause of decline. Use the exit ticket (3 questions): (a) Calculate the estimate when 30 are marked, 40 are caught in the second sample and 8 are marked. (answer 150) (b) Give one assumption of capture-recapture. (c) How could this help the community near the river? Collect and sort into 'secure', 'nearly there' and 'needs reteaching'.
Driving Question Board (DQB) Creation	Learners return to the Driving Question Board with the question: 'Why are fish disappearing from a water body that once supported	Say: 'Today we found a way to put a number on the fish. Which questions on our board did this help answer? What new questions	Driving Question Board: Public tracking of questions and answers helps learners see their progress, link each lesson to the anchoring	Indicators: (1) at least half of the learners contribute a note; (2) new questions refer to causes of population change (biotic and

 VIDEO: 6a-c. Population ecology <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Population Size, Density, and Distribution (Flexbook) Source: CK-12  Search ARES for similar readings	many fish, and what can the community do to restore the ecosystem?' On sticky notes they add what they have learned (e.g. 'We can estimate fish numbers using capture-recapture') and new questions (e.g. 'Why would the number of fish fall? Does water pollution, overfishing or too many weeds change the number? How many fish can the river support?'). They place notes under categories such as 'Measuring populations', 'Causes of decline', 'Possible solutions'. They mark which earlier questions were partly answered by the simulation.	do you have?' Give WAIT TIME of 10-15 seconds before taking responses. Cold-call 4 learners to read and place a note. Highlight questions on factors that change population size: 'Several of you are asking why numbers change. That is exactly where we are going next.' Ask: 'Who in our community could use this number, and for what?' (e.g. Beach Management Unit, fisheries officers, county government).	phenomenon, and build ownership of the investigation. Categorising questions shows how the measurement skill leads to the question of causes.	abiotic factors), not only repeat earlier ones; (3) learners can state which earlier question was partly answered. Record the questions to plan Lesson 7 on factors affecting population size.
Model Building Phase  VIDEO: Worked example: Logistic model word problem <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Population Size, Density, and Distribution Practice Source: CK-12  Search ARES for similar readings	In their groups, learners revise their class model of the river ecosystem (a diagram showing fish, algae, aquatic weeds, bacteria, water, farms and homes upstream). They add a new section: 'Measuring the fish population', showing how the capture-recapture method is carried out in the river (netting, tagging, releasing, recapturing) and the formula. They use an arrow and label to show where a decline in estimates (e.g. 500 to 250) would sit in the model, with a question mark for the causes still unknown. Groups add a 'What we still need to find out' box: 'What factors affect the population size of fish in the river?' Each group shares its revised model in a quick gallery walk and gives one piece of feedback to another group.	Say: 'Update your model. Where does counting fit in? What can it tell us and what can it not tell us?' Give WAIT TIME of 10-15 seconds before groups begin drawing. Move around and ask: 'Where would you put the decline in fish numbers? What might be linked to it?' Cold-call 3 groups in the gallery walk to explain one change they made and why. Remind learners to use resources prudently (reuse paper, share markers). Close by telling the class: 'We can now estimate how many fish there are. Next lesson we ask: what factors affect population size in an ecosystem?' Distribute no homework other than reflecting on that question.	Cumulative model revision: Adding the measurement method and unanswered questions to the model shows learners how new evidence refines their explanation. Gallery-walk feedback encourages learners to justify their model with evidence.	Check that: (1) the model includes the capture-recapture steps and formula; (2) the decline in numbers is linked to possible causes with a question mark rather than an unsupported claim; (3) learners give feedback that uses evidence from the simulation; (4) the next-step question about factors affecting population size is stated clearly.

D. TEACHER REFLECTION

After the lesson, consider: Did every learner have a turn in each jigsaw role? Did groups mix the counters well, and how did this affect their estimates? Which learners

still think an estimate must be exactly equal to the true count? Were the Lake Victoria calculations accessible, or do some learners need more support with ratios? How well did the exit ticket show who can apply the formula and state assumptions? How can the questions on the DQB be used to plan Lesson 7 on factors affecting population size? What would you change about timing or materials (e.g. using beans or bottle tops) next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We marked 20 of 100 counters, mixed them, took a second sample of 20 and counted how many were marked (e.g. 5). Estimates varied between trials but were near the true value of 100.
What did I learn?	Population size can be estimated by capture-recapture using: $\text{estimated population} = (\text{number initially marked} \times \text{number in second sample}) / \text{number recaptured}$. More trials give more reliable estimates, and the method relies on assumptions such as thorough mixing and no migration, births or deaths.
How does this explain the phenomenon?	We can estimate the number of fish in the river without counting every fish, and compare estimates over time to show whether the population is falling. The method shows how many fish there are but not why they are decreasing, so we must investigate the factors that affect population size next.

LESSON 7 (80 minutes (2 x 40-minute periods)): Factors Affecting Population Size: Why Are the Fish Disappearing?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use a Khan Academy reading and a PhET predator and food scarcity simulation to identify biotic and abiotic factors that affect population size, and use this evidence to explain why fish numbers are dropping in the Kenyan river while algae and weeds boom.
Knowledge	Learners describe factors affecting population sizes in ecosystems (KICD SLO c): biotic factors (predation, competition, disease, food availability) and abiotic factors (temperature, dissolved oxygen, light, water quality, nutrients). They explain carrying capacity and limiting factors.
Skills	NGSS SEP: Analyzing and Interpreting Data (reading population graphs from the simulation); Planning and Carrying Out Investigations (changing one variable at a time in the simulation); Engaging in Argument from Evidence (ranking factors in the discussion diamond); Constructing Explanations.
Attitudes	Core competencies: Self-efficacy (communicating clearly while debating causes) and Citizenship (solving problems in the local community). Values: Integrity (using resources prudently) and Patriotism (caring for Kenya's rivers and lakes). PCIs: Waste Management and Prevention of Global Warming.
Key Inquiry Question	How do organisms interact with their environment?
Purpose in Storyline	Lessons 1 to 6 built the vocabulary of ecology, adaptations, ecosystems and population estimation. Learners now see that population size is controlled by interacting biotic and abiotic factors. This explains why fish decline while algae boom, but not yet why nutrient inputs trigger the boom. That gap leads to Lesson 8: How do nitrogen and carbon affect living organisms?
Safety Notes	Computer/tablet use: share devices in groups of 3 to 4, and keep cables tidy and away from walkways. If the offline platform is used, load the simulation and reading before class. No water samples are handled in this lesson. If a river visit is mentioned, remind learners never to go near rivers without an adult.

B. LESSON OVERVIEW

Learners return to the video of the Kenyan river (2:20 to 4:50) in which fish are declining while algae, weeds and a bad smell increase. They predict which factors could reduce fish numbers, read a Khan Academy passage on limiting factors and population growth, and manipulate the PhET simulation to see how predators and food scarcity change population size. In a discussion diamond they rank nine factors by how well each explains the river's fish decline, using evidence. They add new questions to the Driving Question Board and revise their model. The lesson ends with a puzzle: if too many nutrients feed algae, how do nitrogen and carbon move through living things and affect them?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 01 Biotic And	Learners replay the video clip (2:20 to 4:50) of the river with fewer fish, more algae, weeds and	Say: 'Watch again. This time, look for anything living or non-living that could change how many fish	Prediction with sketch graph and sorting: committing to a prediction reveals prior ideas, so later	Look for: (1) each learner has two written factors and a sketch graph showing a downward trend; (2) at

Abiotic Factors Source: Seavuria Search ARES for similar videos  HTML: Limiting Factors to Population Growth (Flexbook) Source: CK-12 Search ARES for similar readings	a smell. Individually, in 3 minutes, they write on a sticky note or in their notebooks: 'Which TWO things do you think are reducing the number of fish (e.g. tilapia, mudfish, omena) in this river? Draw a quick sketch graph of fish numbers over 20 years.' They then pair-share and sort their ideas into 'living things' and 'non-living things'. Their earlier ideas (fish are old, no food) are revisited.	survive.' Play the clip once. Say: 'Write down two factors and sketch a graph. You have 3 minutes.' Circulate and note common ideas such as overfishing, pollution, predators and low oxygen. After pairs share, say: 'Think about this for a moment,' and give WAIT TIME of 10-15 seconds before asking: 'Which of your factors is living and which is non-living?' Cold-call 4 learners from different groups. Record the two categories on the board as 'Biotic' and 'Abiotic' without correcting yet. Ask: 'Our earlier ideas did not explain the algae and the smell. Do your new factors help?'	evidence can confirm or challenge them. Sorting into living and non-living prepares the biotic/abiotic vocabulary.	least 3 of 4 cold-called learners can place a factor in the correct category; (3) some learners link a factor to algae or smell. Note learners who only list 'old fish' or 'no food' and revisit them in the Explain phase.
Observe Phase  VIDEO: 01 Biotic And Abiotic Factors Source: Seavuria Search ARES for similar videos  EXERCISE: Limiting Factors to Population Growth Practice Source: CK-12 Search ARES for similar readings	Part A (15 minutes): In groups of 4, learners read the Khan Academy passage on population growth and limiting factors (link: reading here). Each learner highlights: (a) a biotic factor, (b) an abiotic factor, and (c) the meaning of carrying capacity. Groups complete a two-column table with Kenyan examples, e.g. Nile perch preying on small fish in Lake Victoria, competition among tilapia for food, drought lowering water levels in the Tana River, and warm water holding less oxygen. Part B (25 minutes): Using the PhET simulation (link: SIMULATION HERE), learners run three trials: (1) a baseline with fish and food only, (2) adding predators, and (3) reducing food. They record population counts at set time intervals and sketch the graphs. Each trial changes only one variable. They also note what the simulation cannot show, such as algae, oxygen and smell.	Distribute devices and reading. Say: 'Annotate as you read. Find one biotic factor, one abiotic factor, and the meaning of carrying capacity.' Circulate and ask: 'Where in the river video do we see this factor?' For the simulation say: 'Predict first. If we add predators, what will happen to the prey graph? Write it down before you click.' Allow WAIT TIME of 10-15 seconds after each prediction prompt. Remind: 'Change only ONE thing at a time, otherwise we cannot tell what caused the change.' Ask groups: 'What happened to the food when prey numbers were high?' Cold-call 3 groups to report their trial 2 and trial 3 results. Prompt: 'What does this simulation leave out that the river video shows?' (Expected: algae, oxygen, pollution, fishing by people.) Check that devices are shared fairly.	Evidence gathering from two sources (text and simulation): the reading provides vocabulary and concepts, while the simulation lets learners test cause and effect by controlling variables. Noticing the simulation's limits builds model-critique skills.	Observable indicators: (1) the table has at least 2 correct biotic and 2 correct abiotic factors with Kenyan examples; (2) data tables show population counts for all three trials; (3) graphs show prey rising and falling with predator changes, and a decline with low food; (4) learners state that only one variable was changed per trial. Oral check: 'What is carrying capacity?' Look for 'the maximum population the environment can support'.
Explain Phase	Learners run a Discussion	Say: 'You have nine factor cards.'	Discussion Diamond: ranking	Oral questions (as planned by the

 VIDEO: Ecology introduction <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Population Size Practice Source: CK-12  Search ARES for similar readings	Diamond in groups of 4 (9 cards in a diamond shape: 1 top, 2, 3, 2, 1 bottom). The nine factor cards are: overfishing by people, predation (e.g. Nile perch), competition for food, disease and parasites, low dissolved oxygen, high water temperature, excess nutrients from farm fertiliser and sewage, reduced light from algae cover, and drought lowering water level. Groups rank them from 'most important in explaining the river's fish decline' (top) to 'least important' (bottom), citing evidence from the video, the reading and the simulation. They write one sentence on each of their top three: 'Factor ____ reduces fish because ____, which is shown by ____.' They then compare diamonds with another group and defend their choices. Finally, each learner writes a 3-sentence claim-evidence-reasoning response to: 'Why are fish disappearing while algae increase?'	Place the one that best explains our river at the top and the weakest at the bottom. Every placement needs evidence.' Allow 12 minutes. Circulate with probes: 'Which evidence supports putting this at the top?' 'Does this factor explain the algae AND the smell?' 'Is it biotic or abiotic?' Use WAIT TIME of 10-15 seconds after each probe. Hold a gallery walk: groups visit one other diamond and ask one question. Cold-call 4 groups to justify their top and bottom cards. Push: 'Could one factor cause another? For example, how might algae cause low oxygen?' Draw links on the board between nutrients, algae, light, oxygen and fish. Say: 'Notice that no single factor works alone. Factors interact.' Correct misconceptions, such as 'algae are bad because they are poisonous' versus 'when algae die and decompose, oxygen falls'.	forces learners to weigh evidence, compare competing explanations and argue from evidence. It shows that factors interact and that no single cause fully explains the phenomenon. Claim-evidence-reasoning writing consolidates individual understanding.	teacher): 'What is the difference between a biotic and an abiotic factor?' 'Why can a population not grow forever?' 'How might excess nutrients lead to fewer fish?' Success indicators: learners use the terms biotic, abiotic, limiting factor and carrying capacity correctly; they cite evidence for rankings; at least 3 of 4 cold-called groups link nutrients, algae and oxygen. Note learners who rank by opinion without evidence and prompt them with 'What did you see in the data?'
Driving Question Board (DQB) Creation  VIDEO: Worked example: Logistic model word problem <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Limiting Factors to Population Growth	Learners revisit the class Driving Question Board with the question: 'Why are fish disappearing from a water body that once supported many fish, and what can the community do to restore the ecosystem?' Groups write on sticky notes (a) one question that has been partly answered by today's work, with the answer, and (b) one new question that has arisen, e.g. 'Where do the extra nutrients come from?', 'How do nitrogen and carbon move through plants, algae and fish?', 'How can farmers around Lake Victoria reduce runoff?'. They place the notes on the board under the headings 'Answered', 'Partly answered' and 'New'.	Say: 'Look at our board. Which questions did today's evidence help us answer?' Allow WAIT TIME of 10-15 seconds. Invite 3 learners to move a note to 'Answered' or 'Partly answered' and justify it. Say: 'Write one new question that today's evidence raised.' Cold-call 3 groups to read their new question. Highlight those about nutrients, nitrogen and carbon. Say: 'Many of you are asking where the extra nutrients come from and how nitrogen and carbon move. That is our next investigation.' Group similar questions and keep the board visible.	DQB tracking: moving and adding questions makes growing understanding visible and gives learners ownership of the investigation. It shows how today's evidence leads to the next lesson.	Indicators: each group contributes one answered/partly answered note and one new note; at least 2 new questions refer to nutrients, nitrogen, carbon or human activity; learners justify the placement of a note using evidence from the reading or simulation.

(Flexbook) Source: CK-12 Search ARES for similar readings				
Model Building Phase  VIDEO: The logistic growth model <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Limiting Factors to Population Growth Practice Source: CK-12 Search ARES for similar readings	Learners revise their ecosystem model of the river (started in earlier lessons) on chart paper. They add: (1) biotic factors (fish, predators such as Nile perch or birds, competitors, algae and weeds, decomposers), (2) abiotic factors (temperature, light, dissolved oxygen, nutrients, water level), (3) arrows showing relationships, e.g. 'excess nutrients, more algae, less light and oxygen, fewer fish', and (4) human activity: fishing, fertiliser from farms in the catchment, and untreated sewage or waste. They mark with a '?' the parts they cannot yet explain, such as how nitrogen and carbon move through the system. Each group writes a one-sentence prediction for how restoring the river might change the fish graph.	Say: 'Open your river model. Add the factors we ranked today and draw arrows to show how they affect fish numbers.' Circulate and ask: 'What does this arrow mean?' 'Which factor here is a limiting factor?' 'Where did the evidence come from?' Use WAIT TIME of 10-15 seconds. Insist on labelled arrows and units where possible. Say: 'Put a question mark where your model cannot yet explain what is happening. Next lesson we will investigate nitrogen and carbon.' Cold-call 3 groups to share one new feature of their model and one gap. Remind learners to use their materials prudently (reuse paper, share markers) to show integrity. Close by posing: 'How do nitrogen and carbon affect living organisms?'	Cumulative model revision: adding factors, relationships and gaps turns separate pieces of evidence into a coherent causal explanation. Marking unknowns identifies what to investigate next.	Indicators: the model includes at least 3 biotic and 3 abiotic factors; at least 3 labelled arrows show cause and effect; human activities are linked to nutrient input; at least one '?' marks a gap related to nitrogen or carbon; groups can explain one arrow when asked. Use these to plan Lesson 8.

D. TEACHER REFLECTION

Which learners could separate biotic from abiotic factors and use evidence in their rankings? Which still rely on one cause such as overfishing? Did the simulation help learners see that factors interact, and did they notice what it left out (algae, oxygen, nutrients)? Were the discussion diamond rankings supported by evidence or opinion? Which new DQB questions about nitrogen and carbon should be used to open Lesson 8? Were devices shared fairly, and what timing adjustments are needed?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that in the PhET simulation adding predators made the prey population fall and then rise and fall again, and reducing food made the population decline. We also read about biotic and abiotic limiting factors and saw in the river video that fish decreased while algae, weeds and a bad smell increased.
What did I learn?	We learned that population size is controlled by biotic factors (predation, competition, disease, food) and abiotic factors (temperature, oxygen, light, nutrients, water level). Populations stop growing at the carrying capacity, and factors interact with each other.
How does this explain the	Fish are probably disappearing because several factors work together: overfishing, predators and competition, plus extra nutrients

phenomenon?	that cause algae and weed growth, which reduces light and dissolved oxygen. The simulation cannot explain where the nutrients go, so we need to investigate how nitrogen and carbon affect living organisms.
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LESSON 8 (80 minutes (2 x 40-minute periods)): Nitrogen and Carbon Cycles: How Nutrients Move and Why Algae Boom in Our River

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners illustrate the nitrogen and carbon cycles in ecosystems and use them to explain how nutrient enrichment from human activity leads to algae and weed growth, bad smells and fish loss in the river from the anchoring phenomenon.
Knowledge	Illustrate the nitrogen and carbon cycles in ecosystems (KICD SLO e). Describe the roles of nitrogen-fixing bacteria (e.g. Rhizobium in bean and groundnut roots), decomposers, producers and consumers in the cycles. Link excess nitrates and phosphates from fertiliser, livestock waste and sewage to algae growth.
Skills	Developing and using models (NGSS SEP 2); constructing explanations; analysing and interpreting information; communicating clearly with peers through annotated diagrams and peer review.
Attitudes	Integrity: utilises resources prudently while making charts and models to illustrate nitrogen and carbon cycles. Self-efficacy: communicates clearly with others. Citizenship: concern for solving problems in the local community. Appreciation of the link between environmental pollution and climate change.
Key Inquiry Question	How do organisms interact with their environment?
Purpose in Storyline	Lessons 1-7 showed that organisms are adapted to habitats and that populations depend on biotic and abiotic factors. Lesson 8 supplies the missing mechanism for the extra algae, weeds and smell in the river: nutrients cycle between living things and the environment, and human activity can overload them. It ends with the new question: Does pollution affect organisms?
Safety Notes	Use only chart paper, markers, string, sticky notes and pre-printed cards. If learners visit a river or pond, they must not touch or drink the water and must stay with the teacher. Wash hands after handling any soil or plant samples. Keep walkways clear during gallery walks.

B. LESSON OVERVIEW

Learners return to the Kenyan river video (2:20 to 4:50) and predict why algae, weeds and smell increased while fish decreased. They then discuss the nitrogen and carbon cycles using Kenyan examples (maize shamba with beans, a dairy farm near Kericho, sewage and farm runoff entering a river flowing to Lake Victoria). In groups they build an annotated diagram (model) of both cycles, adding a pathway where excess nutrients enter the river. Groups then do peer review using a checklist. Learners update the Driving Question Board and the class model, and finish with the new question: Does pollution affect organisms?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: The nitrogen cycle Source: Khan	Learners re-watch the river video clip (2:20 to 4:50) in 3 minutes. On a sticky note they individually write a prediction: 'Where do algae and	Play the clip and say: 'Watch the river again. Look for clues about what is entering the water and what is growing in it.' Pose: 'Algae	Prediction with partner talk (Think-Pair-Share): surfacing prior ideas about nutrients and growth creates a need to know and gives a	Listen for: learners mention fertiliser, manure, sewage or soap (strong); learners only mention 'sunlight' or 'old fish' (needs

Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Nitrogen Cycle Practice Source: CK-12 Search ARES for similar readings	weeds get the materials to grow so much, and could this link to fewer fish?' They sketch a quick arrow diagram of where they think extra nutrients come from (e.g. maize farms using fertiliser, cattle waste, latrines, washing detergents). They share with a partner, then with the class.	and weeds are growing fast here. What do plants need to grow fast?' WAIT TIME 10-15 seconds in silence before taking answers. Cold-call 4 learners to share predictions without judging, e.g. 'Akinyi, what did you write? Kamau, do you agree or want to add?' Record all predictions on the board under 'Our Ideas Before'. Ask: 'In Lesson 1 some of you said the fish are old or have no food. Does that still explain the algae?'	baseline to compare against after the model is built.	probing). Tally on a checklist how many name a human source of nutrients. Look for sketches with arrows from land to river.
Observe Phase VIDEO: The nitrogen cycle Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Nitrogen Cycle (Flexbook) Source: CK-12 Search ARES for similar readings	Learners read the lesson resource (link/book page provided by the teacher) and take part in a guided discussion on the nitrogen and carbon cycles. In groups of 4 they study cycle cards: nitrogen gas in air, nitrogen fixation by Rhizobium in bean/groundnut root nodules and by lightning, ammonium, nitrification (nitrifying bacteria), nitrates absorbed by maize and sukuma wiki, proteins eaten by cattle and people, death and decomposition, denitrification; and for carbon: CO₂ in air, photosynthesis, respiration, feeding, decomposition, combustion of firewood and fossil fuels, dissolved CO₂ in water. They then make a model: a large annotated diagram of both cycles on chart paper using arrows, labelling each process and organism, with Kenyan examples (e.g. githeri crops, dairy cows in Kericho, Lake Victoria, charcoal burning). They add an extra arrow showing fertiliser and sewage runoff entering the river.	Introduce: 'Today we follow the journey of nitrogen and carbon through our shamba and our river.' Facilitate the discussion with questions: 'Why do farmers in Kisii and Bungoma plant beans between maize rows?' 'Where does the carbon in your ugali end up when you run?' Give each group a set of cycle cards and chart paper; remind them 'use resources prudently, one chart per group, reuse scrap paper.' Circulate and ask each group: 'Which arrow shows how nitrogen gets from air into beans? What happens to the nitrogen when the cow dies?' Cold-call 3 groups to explain one process. Use WAIT TIME 10-15 seconds after each question. Prompt: 'Where could extra nitrates enter your diagram from the farm, latrine or sewer? Draw that arrow in a different colour.' Support groups who confuse nitrification and denitrification with sentence frames.	Annotated diagram: learners build a labelled model with explanatory notes beside each arrow (what happens, which organism does it, why it matters). Writing annotations forces them to articulate mechanisms rather than just copy arrows, and the added 'pollution' arrow links the cycles to the phenomenon.	Observable indicators: (1) both cycles drawn as closed loops with correct arrow direction; (2) at least 5 processes correctly named in each cycle; (3) decomposers and bacteria shown as key organisms; (4) an extra input arrow showing nutrient runoff labelled in a different colour; (5) annotations in own words. Use a tick checklist while circulating.
Explain Phase VIDEO: The nitrogen cycle	Groups use their annotated diagrams to write a 4-5 sentence explanation answering: 'How can too much nitrogen (nitrates) and	Say: 'Use your diagram as evidence. Point to the arrow that explains the algae.' Give the chain scaffold on the board with blanks.	Peer review with a criteria checklist plus Claim-Evidence-Reasoning: learners test their model against others' ideas, justify	Observable indicators: explanation includes nutrient source, algal bloom, oxygen depletion by decomposers, and fish death; peer

Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: The Nitrogen Cycle (Flexbook) Source: CK-12 Search ARES for similar readings	phosphates entering the river lead to more algae, a bad smell and fewer fish?' They sequence the chain: fertiliser/manure/sewage runoff, nutrient enrichment, algal bloom, light blocked for submerged weeds, algae and weeds die, decomposers (bacteria) multiply and use up dissolved oxygen in respiration, fish suffocate, anaerobic decay produces foul smell. Then groups do peer review: swap diagrams with another group, use a checklist, and give 'two stars and a wish' feedback. They revise their own diagram after feedback.	Pose: 'If decomposers use oxygen, what happens to fish that need oxygen?' WAIT TIME 10-15 seconds. Hand out the peer review checklist (correct arrows, labelled processes, link to river, clarity, Kenyan example) and say: 'Be kind, be specific, be helpful. Say what is correct, then one thing to improve, with a reason.' Cold-call 3 reviewers to report one piece of feedback they gave and 2 authors to say what they will change. Correct misconceptions: plants do not 'eat' nitrogen gas directly; carbon cycle and nitrogen cycle are linked through decomposition and respiration.	their explanation using the diagram as evidence, and refine it, which deepens understanding of cause-and-effect chains.	feedback is specific and refers to the checklist; learners revise at least one element after feedback. Note learners who still say 'fish have no food' without mentioning oxygen.
Driving Question Board (DQB) Creation VIDEO: The nitrogen cycle Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Nitrogen Cycle in Ecosystems (Flexbook) Source: CK-12 Search ARES for similar readings	Learners return to the Driving Question Board. They place sticky notes answering earlier questions (e.g. 'Why is there so much algae?') with evidence from today. They write new questions, e.g. 'Do chemicals from factories and plastics also harm fish?', 'Does the pollution affect the fish we eat, and people?', 'How does carbon dioxide link to global warming?', 'What can our community do to stop runoff?' They group questions into categories: causes, effects, solutions.	Say: 'Which questions on our board can we now answer? Move them to the Answered column and show the evidence.' Ask: 'What new questions has today raised?' WAIT TIME 10-15 seconds. Cold-call 4 learners to place and read a question. Highlight a question about pollution and organisms and say: 'Our cycles show nutrients can be overloaded. Does pollution affect organisms? That is our next lesson.' Ensure question wording is clear and testable.	Driving Question Board tracking: sorting answered and unanswered questions shows learners their growing understanding and builds ownership of the investigation sequence.	Look for questions that refer to cycles, oxygen, nutrients or pollution rather than vague ones. Count how many learners add a new question and how many cite evidence from today's diagram for an answered question.
Model Building Phase VIDEO: The nitrogen cycle Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners update the class model of the river ecosystem. On the class chart, they add nutrient cycling: producers (algae, water weeds), consumers (fish, tilapia, insects), decomposers, and the pathways of nitrogen and carbon, plus human inputs (fertiliser, manure, sewage, detergents).	Say: 'Our model from Lesson 7 showed factors affecting populations. What must we add today?' Invite 3 learners to add components to the class model and explain the arrows. Cold-call 3 more to state one thing in the model that has changed since Lesson 1. Use WAIT TIME 10-15	Cumulative model revision: adding nutrient cycles to the existing river ecosystem model helps learners integrate new evidence into a coherent explanation and identify gaps that motivate the next lesson.	Check notebook models for: nutrient input arrow, algae growth, oxygen decline, fish decline, and CO₂ from respiration. Use a 3-level rubric (complete, partial, beginning). Ask 2 learners to explain their model in one minute to check clarity.

for similar videos  EXERCISE: Nitrogen Cycle in Ecosystems Practice Source: CK-12  Search ARES for similar readings	Individually they add to their notebook a revised model showing why fish numbers drop, with at least one arrow for nutrient input, one for oxygen use and one for CO2 release. They complete the summary table.	seconds. Ask: 'What does our model still not explain?' (e.g. toxic chemicals, plastic, heat). Close with: 'We have explained the algae and smell. But does pollution affect organisms in other ways? Keep that question for our next lesson.' Remind learners to reuse materials and tidy up.		
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D. TEACHER REFLECTION
Did learners correctly place arrows and processes in both cycles, or did they confuse nitrification and denitrification? Which groups linked nutrient enrichment to oxygen depletion and not just to 'more algae'? Was the peer review feedback specific and useful? How well did learners use Kenyan examples such as Rhizobium in beans, farms near Kericho and runoff into Lake Victoria? What adjustments are needed to wait time, grouping, or resources before Lesson 9 on the effects of pollution?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that the nitrogen and carbon cycles move nutrients between air, soil, water and living things through fixation, nitrification, photosynthesis, respiration and decomposition, and that runoff of fertiliser, manure and sewage adds extra nutrients to the river, with more algae, weeds and bad smell.
What did I learn?	We learned that decomposers and bacteria recycle nutrients, that too many nutrients cause algal bloom (eutrophication), and that when algae and weeds die, decomposers use up dissolved oxygen, which harms fish.
How does this explain the phenomenon?	The river is losing fish because nutrient pollution from farms and settlements causes algae and weeds to boom; their decay by bacteria uses up oxygen and produces foul smell, so fish suffocate. New question: Does pollution affect organisms?

LESSON 9 (80 minutes (2 x 40-minute periods)): Pollution, Garbage Rivers and Restoring Our River: The Final Explanation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To use evidence from a pollution walk in the neighbouring shopping centre, and research on garbage rivers and effluent in the Nairobi and Athi Rivers, to write the Final Explanation of why fish are disappearing while algae and weeds boom and the water smells, and to propose community solutions to restore the river.
Knowledge	By the end of the sub strand, the learner should be able to: describe the effects of pollution in the environment (including global warming and climate change), and use the nutrient cycles, population factors and interactions from Lessons 1-8 to explain the anchoring phenomenon.
Skills	Developing and using models (comparing the Lesson 1 model with the final model); Obtaining, evaluating and communicating information (walk observations, internet research, presentations); Constructing explanations and designing solutions (Final Explanation and community action plan); Analysing and interpreting data (classifying pollution types and linking them to effects).
Attitudes	Citizenship: solves problems in the local community by making posters to sensitise the community about the effects of environmental pollution. Self-efficacy: communicates clearly with others while discussing the causes and impacts of pollution and climate change. Integrity: utilises resources prudently. Patriotism: demonstrates love for the country while sensitising the community about the effects of environmental pollution. Waste Management PCI: appreciates the importance of proper waste management. Prevention of Global Warming PCI: appreciates the need to reduce greenhouse gas emissions.
Key Inquiry Question	How do organisms interact with their environment?
Purpose in Storyline	This is the final lesson. Learners have studied ecosystems, adaptations, population estimation, limiting factors and nutrient cycles. Now they connect human activity (pollution) to the disappearing fish, algae and weed growth and the smell in the Kenyan river video (2:20 to 4:50), write the Final Explanation, compare the Lesson 1 model with the final model, and complete the DQB. Their new question for further study: how do garbage rivers and effluent in the Nairobi and Athi Rivers continue to affect ecosystems and people downstream?
Safety Notes	Walk: brief learners on road safety, keep them in pairs or groups of four with a teacher at the front and back, and stay on the pavements. Do NOT touch litter, broken glass, sewage, or burning waste; observe and record only, and use gloves if collecting any sample. Wash hands with soap after returning. Avoid inhaling smoke from burning rubbish. For internet research, use school-approved devices or printed extracts if offline.

B. LESSON OVERVIEW

Learners begin by predicting which kinds of pollution they will find in the neighbouring shopping centre and how they might end up in a river. They then walk to the shopping centre in groups to record the types of pollution (solid waste such as plastic bottles and polythene, air pollution from matatus and boda bodas and burning rubbish, noise, oil and detergent runoff from car washes, and blocked drains). Back in class, they research and present findings on the effects of pollution, focusing on garbage rivers and effluent in the Nairobi and Athi Rivers and on global warming and climate change. In the Explain phase they write the Final Explanation for the river phenomenon: effluent and fertiliser runoff add nitrates and phosphates, causing algal and weed growth, bacterial decomposition removes dissolved oxygen, fish die or leave, and decomposing matter produces the smell. They complete the DQB, compare the Lesson 1 model with the final model, and propose community solutions. The lesson closes with an exit ticket.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: ecology pollution effluent eutrophication  HTML: Effects of Water Pollution (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the river video (2:20 to 4:50) shown in Lesson 1: fewer fish, more algae and weeds, and a bad smell. Individually, they write a prediction in their notebooks: 'Which types of pollution will I find at the shopping centre, and how could each reach a river like the Nairobi River?' They share with a partner and draw a quick arrow sketch from the shopping centre to the river, then predict the effect of each pollutant on fish.	Display the phenomenon image and say: 'Remember the river that once had many fish. Today we look for what humans may be adding to it.' Give 3 minutes of silent writing. Then say: 'Turn to your partner and share one pollutant you expect to see.' Use WAIT TIME of 10-15 seconds after asking: 'Which of these pollutants could make algae and weeds grow faster?' Cold-call 4 learners from different groups and record predictions on the board under headings: Solid, Air, Water, Noise. Do not correct predictions yet. Give safety briefing and assign roles: recorder, observer, photographer/sketcher, timekeeper.	Prediction with partner sharing and a drawn pathway: surfacing prior ideas about sources and pathways of pollution lets learners test them against real evidence on the walk, and creates a cognitive need to look for links between pollution and the river.	Observable indicators: at least 3 pollutant types appear in each learner's prediction; learners draw an arrow from the shopping centre (drains, runoff) to the river; learners who mention only litter are prompted to consider air, oil, detergent and sewage pollution. Teacher tallies the most common predictions.
Observe Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: ecology pollution effluent eutrophication  EXERCISE: Effects of Water Pollution Practice Source: CK-12  Search ARES for similar readings	Learners walk in groups to the neighbouring shopping centre and note every type of pollution they observe: plastic bottles and polythene bags, food waste from kiosks selling mandazi and roasted maize, smoke from burning rubbish, exhaust from matatus and boda bodas, noise from hooting and loudspeakers, oily puddles near garages, soapy water from car washes flowing into drains, and blocked drains. They fill an observation table (type of pollution, location, probable source, where it may end up) and sketch or photograph the drains. Back in class, groups use the internet (or printed extracts) to research the effects of pollution, with each group taking one focus: (1) garbage rivers and plastic, (2) effluent and sewage in the Nairobi	Before leaving say: 'You are scientists collecting evidence. Observe and record only, and do not touch the waste.' Circulate and ask: 'Where do you think this water goes when it rains?' and 'What is it carrying?' WAIT TIME of 10-15 seconds before accepting answers. Back in class say: 'Use at least two sources and note where the information comes from.' Check that sources are credible (NEMA, UNEP, Kenyan news or university sources). Cold-call 2 members from each group to explain one finding in their own words before presentations. Prompt: 'How does your focus connect to the algae, weeds, smell and missing fish in our river?' Have each group present while others fill a summary table.	Evidence-gathering from local fieldwork and research, followed by group presentations, so learners connect observed local pollution to documented effects. Class-built cause-and-effect chains (for example, fertiliser, detergent or sewage leads to nitrates and phosphates, leading to algal bloom, bacterial decomposition, low dissolved oxygen and fish death) make hidden mechanisms visible.	Observable indicators: observation tables list at least 5 pollution types with a source and likely destination; groups cite at least two sources; presentations include a cause-and-effect chain; peers can restate the key effect in one sentence. Teacher notes misconceptions such as 'polythene breaks down quickly' or 'only factories pollute' and addresses them in the next phase.

	and Athi Rivers, (3) fertiliser and detergent runoff and eutrophication, (4) air pollution, greenhouse gases, global warming and climate change. Groups prepare a 3-minute presentation with a chart.			
Explain Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: ecology pollution effluent eutrophication  HTML: Effects of Air Pollution on the Environment (Flexbook) Source: CK-12 Search ARES for similar readings	Learners write the Final Explanation for the phenomenon in groups of four, using a claim-evidence-reasoning frame: Claim (why fish are disappearing while algae and weeds boom and the water smells); Evidence (walk observations, research on the Nairobi and Athi Rivers, and earlier lesson evidence on food chains, limiting factors, population estimates and the nitrogen and carbon cycles); Reasoning (nitrates and phosphates from effluent, fertiliser and detergents cause algal and weed growth; dead algae are decomposed by bacteria that use up dissolved oxygen; fish suffocate or leave; anaerobic decomposition produces foul-smelling gases; garbage and toxic chemicals harm and kill fish directly; warmer water from climate change holds less oxygen). They also link burning waste and fossil fuels to greenhouse gases, global warming and climate change. Finally, each group proposes three community solutions, such as proper waste separation and collection, river clean-ups, riparian buffer planting, treating effluent, reducing fertiliser and detergent use, banning single-use plastics, and community posters.	Say: 'Using all the evidence we have collected in nine lessons, explain why our river lost its fish.' Display the CER frame. Give 10 minutes for writing, then WAIT TIME of 10-15 seconds before asking: 'Which piece of evidence best supports your claim about the smell?' Cold-call 3 groups to read their claim and one piece of evidence. Probe: 'What happened to the oxygen, and how do you know?' and 'How is a garbage river in Nairobi connected to climate change?' Ask different groups to challenge each other respectfully: 'Which part of this explanation needs stronger evidence?' Support learners who struggle with sentence starters such as 'Because nitrates and phosphates increased, ...'. Link to values: 'Which actions can our school and community take, and who can lead them?'	Claim-Evidence-Reasoning (CER) group writing with peer challenge: learners combine evidence from all lessons into one coherent causal explanation, justify it with data, and propose solutions, which shows how organisms interact with each other and with human activity in the environment.	Observable indicators: the explanation includes the full chain (nutrient input, algal bloom, bacterial decomposition, low oxygen, fish loss, smell); at least two pieces of evidence from different lessons are cited; the nitrogen cycle and carbon cycle are correctly linked to pollution and climate change; solutions are specific and match the causes. Use a checklist while circulating and give feedback on gaps.
Driving Question Board (DQB) Creation	Learners revisit the DQB started in Lesson 1. Groups take sticky notes with the questions posed, decide which have been answered	Say: 'Let us look at our questions from Lesson 1. Which have we answered, and what is our evidence?' Give each group 5	DQB sorting and evidence-based answering: tracking progress against the original questions makes learning visible, shows that	Observable indicators: learners place questions in the correct category and cite lesson evidence for each answer; at least three

 VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: ecology pollution effluent eutrophication  EXERCISE: Effects of Air Pollution on the Environment Practice Source: CK-12 Search ARES for similar readings	(with evidence), partially answered, or unanswered, and move them into sections. They write answers on green notes with evidence and place them beside the questions. They add new questions, such as how garbage rivers in the Nairobi and Athi Rivers affect people downstream, how the river can be cleaned, and who is responsible for the effluent.	minutes to sort. WAIT TIME of 10-15 seconds after asking: 'Which question is still unanswered and why?' Cold-call 4 learners to read an answered question and its evidence. Ask: 'What new questions have your findings raised?' Record the new questions and note the next-step question on garbage rivers and effluent in the Nairobi and Athi Rivers. Affirm learners whose questions led to investigations.	science is an ongoing process, and motivates further inquiry.	new relevant questions are added; peers agree or challenge placements with reasons. The teacher records the number of Lesson 1 questions fully answered.
Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: ecology pollution effluent eutrophication  EXERCISE: Water Pollution Practice Source: CK-12 Search ARES for similar readings	Learners take out their Lesson 1 initial model (ideas such as the fish are old or have no food) and draw the final model on a new page or chart. The final model shows: the river ecosystem with fish, algae, weeds, bacteria and decomposers; inputs of effluent, garbage, fertiliser and detergent runoff from the town; nitrate and phosphate enrichment; algal bloom; decomposition and falling dissolved oxygen; fish loss and smell; and air pollution, greenhouse gases and warming. They add labelled arrows, then compare the models in a table (What I thought in Lesson 1 / What I know now / Evidence that changed my mind). Groups then design a poster to sensitise the community about pollution and restoring the river. Finally, learners complete an exit ticket.	Say: 'Place your first model beside the final model. What has changed, and what evidence changed your mind?' Give 10 minutes to draw. WAIT TIME of 10-15 seconds before asking: 'Which part of your first model was missing?' Cold-call 3 learners to explain one revision. Remind: 'Label every arrow with a process, and use resources prudently by reusing paper.' Distribute the exit ticket: (1) Name two types of pollution you observed and one effect of each. (2) Explain in two sentences why fish decline when algae increase. (3) State one action you will take to help restore a river. Collect the tickets as learners leave.	Model comparison (Lesson 1 versus final) and the exit ticket: learners see how their thinking changed, consolidate the full causal explanation in a visual model, and individually show understanding through a final check.	Exit ticket success criteria: names two pollution types with correct effects; explains the algae, oxygen, fish chain correctly; gives a realistic action. Model indicators: includes at least 6 labelled components with correct arrows and shows the changes from Lesson 1. Sort tickets into secure, partial and not yet, and plan follow-up for the next unit.

D. TEACHER REFLECTION

Which learners explained the full chain from nutrient pollution to fish loss without prompting? Which pollution types did the walk reveal that surprised the learners? Did the sources used for research support the claims? Which questions on the DQB remain unanswered, and how will I use the garbage rivers and effluent question in the next unit? How well did the groups collaborate and communicate their ideas? What would I change about the walk's safety arrangements or timing?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed types of pollution at the shopping centre (plastics and polythene, smoke, exhaust fumes, oily and soapy runoff, blocked drains, noise) and found research evidence about garbage rivers and effluent in the Nairobi and Athi Rivers. Add your own observations and photos.
What did I learn?	We learned that pollution adds nutrients and harmful substances to rivers, causing algal bloom, low dissolved oxygen, fish deaths and smell, and that burning waste and fossil fuels adds greenhouse gases that contribute to global warming and climate change. Add what you learned.
How does this explain the phenomenon?	Final Explanation: Fish are disappearing because effluent, garbage and fertiliser runoff enrich the river with nitrates and phosphates, so algae and weeds boom; when they die, bacteria use up the oxygen and produce foul-smelling gases, so fish suffocate or leave. The community can restore the river by managing waste, treating effluent, planting riverbank vegetation and reducing pollution.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.